
LDMX HCal read-out chip board characterization and diagnostics

Author:

Błażej ROZMARYNOWICZ

Supervisor:

Dr. Hannah HERDE

Co-supervisor :

Erik WALLIN



LUND
UNIVERSITY

Division of Particle and Nuclear Physics

Master's Thesis

To fulfil the requirements for the degree of
Master of Science in Physics
at Lund University

May 18, 2026

Abstract

The search for the dark matter (DM) particle has recently found itself at the forefront of physics research. Numerous experiments designed to capture the invisible particle have contributed to constraining its possible energy range. The Light Dark Matter eXperiment (LDMX) aims to delve into the yet unexplored sub-GeV region, searching for “light” DM (LDM). The experiment is designed to infer LDM production from the missing-momentum of electrons, scattered via the “dark bremsstrahlung” process. To identify this event, LDMX employs a wide array of trackers and detectors. Due to the dark bremsstrahlung’s low likelihood, LDMX must diligently exclude all other energy signatures. The hadronic calorimeter (HCal) plays an important role in ensuring the experiment’s high veto power. In this study, we characterize and evaluate the reliability of the HCal’s front-end read-out - the High Granularity Read Out Chip (HGCROC) board. We examine the read-out of ten H2GCROCv3b boards using internal charge-injections. The injections follow a novel automated testing procedure, configured via the `pflib` control software. We review the results over four evaluation metrics to assess the health of the chip board and the control software. Using these metrics, we identify and classify various read-out irregularities. Furthermore, we develop a read-out irregularity identification algorithm, as part of the `pflib` library.

Acknowledgments

I would like to thank my supervisor, Hannah Herde, for her steadfast support and encouragement. Your enthusiasm and passion for physics and teaching made this project truly enjoyable. I can only hope to meet more teachers like you.

I would also like to thank my co-supervisor, Erik Wallin, and Lennart Österman, for their aspirational patience and understanding in showing me the ropes of working with the chip boards. Thank you for answering all my questions.

I want to thank my friends for never letting boredom seep into the PKF division. Special thanks my office-mates, who are an absolute menace.

Finally, I would like to thank my lovely family, who have supported me all these years. Thank you for always helping me pursue my dreams.

Glossary

LDMX Light Dark Matter eXperiment.

SM Standard Model.

(L)DM (Light) Dark Matter.

WIMP Weakly Interacting Massive Particle.

ASIC Application-Specific Integrated Circuit.

HGCROC High Granularity Calorimeter Read Out Chip.

HCal Hadronic Calorimeter.

ECal Electromagnetic Calorimeter.

SiPM Silicon Photo-Multiplier.

CMB Central Mother Board.

(T)DAQ (Trigger and) Data Acquisition.

CC Current Conveyor.

PA Pre-Amplifier.

ADC Analogue-to-Digital Converter.

DAC Digital-to-Analogue Converter.

TDC Time-to-Digital Converter.

TOA Time-Of-Arrival.

TOT Time-Over-Threshold.

bx bunch-crossing.

HR High-Range.

LR Low-Range.

CALIB calibration.

RMSD Root Mean Square Deviation.

Contents

Abstract

Acknowledgements

Glossary

Page

1	Introduction	1
1.1	Overview	1
1.2	Project objectives	1
1.3	Thesis outline	2
2	The Dark Matter Problem	3
2.1	Introduction	3
2.2	Particle Dark Matter	3
2.3	Thermal freeze-out	4
2.4	Light Dark Matter	5
3	The Light Dark Matter eXperiment	6
3.1	Introduction	6
3.2	Experimental design	6
3.2.1	Expected signals & HCal design	8
4	H2GCROCv3: Design and Function	10
4.1	Introduction	10
4.2	Design overview	10
4.3	Digital converters	11
4.3.1	Analogue-to-digital converter	12
4.3.2	Time-to-digital converters	12
4.4	Internal charge-injections	13
4.5	Pflib: control software	13
5	H2GCROCv3b: Characterization	15
5.1	Introduction	15
5.2	Testing setup	15
5.3	Methodology	15
5.3.1	Evaluation metrics	15
5.3.2	Testing procedure	17
5.4	Data analysis	18
5.4.1	Pedestal alignment	18
5.4.2	Peak-ADC alignment	20
5.4.3	ADC/TOT linearity	28
5.4.4	TOA response	31

6	Sample-clustering Identification Algorithm	36
6.1	Introduction	36
6.2	Algorithm versions	36
6.3	Analysis tools	37
6.4	Future improvements	38
7	Conclusion & Outlook	39
	Appendices	41
A	Root-mean-square deviation	41
B	R^2 coefficient	41
C	Testing procedure	41
D	pflib analysis scripts	42
E	Supplementary figures	44
E.1	Pedestal alignment	44
E.2	Peak-ADC alignment	45
F	Pedestal-alignment RMSD threshold	47
G	TOA/TOT triggers and misfires	47
H	Identification algorithm plot examples	50

1 Introduction

1.1 Overview

It would be no exaggeration to call the search for dark matter (DM) the holy grail of astroparticle physics. Even going back as far as the late 19th century, we can find the first attempts to capture DM's essence in Lord Kelvin's research [1]. Over the years, many solutions have been proposed in an attempt to address the problem - an amalgam of theoretical predictions and experimental observations of seemingly invisible gravitational sources. Various theories, ranging from modified gravity to extensions of the Standard Model (SM) of particle physics [2], have been formulated to explain this 'missing mass', which we now believe accounts for 84% of all matter in the Universe [2]. In recent years, hypotheses of particle-DM have experienced a rapid gain in popularity. As a result, numerous experiments were designed and stationed all around the globe, their results consistently chipping away at the predicted energy range of the elusive DM particle.

The Light Dark Matter eXperiment (LDMX) aims to open up the current limits on the so-called "light" region of the DM mass range. To get around DM's characteristic invisibility, LDMX uses missing-momentum techniques. The experiment places trackers on either side of a fixed target to measure the incoming and outgoing momenta of accelerated electrons. Some electrons will scatter off of the target, losing momentum via photons or, rarely, via light dark matter (LDM). The energy signatures of electrons and photons will then be measured by an electromagnetic calorimeter (ECal). In contrast, LDM will deposit no such signature, leaving only an electron with curiously missing momentum to be identified. This allows LDMX to infer LDM production from missing electron momentum. To ensure accurate results, LDMX must diligently account for all other signatures, making its veto power of crucial importance. Thus, the experiment comprises multiple detectors, with a hadronic calorimeter (HCal) accounting for most of its volume.

The HCal's main role is identifying hadrons, which may be produced throughout the experiment. Its front-end read-out is managed by application-specific integrated circuits (ASICs) called the High Granularity Calorimeter Read Out Chips (HGCROCs). While the promise of LDMX's high sensitivity is dependent on the reliability of all its components, the HGCROCs represent an important piece of the puzzle. As such, electrical tests of the chips and their control software mark a fundamental step in preparation for the experiment.

1.2 Project objectives

In preparation for future LDMX operations, the primary objective of this study is characterization of ten chip boards, each one housing an H2GCROCV3b unit. Specifically, this work will delve into the read-out from analogue and time digital converters, responsible for digitizing the raw electrical signals received by the chips. The tests comprise of varied charge-injections following an automated testing procedure, based on existing and novel features of the `pflib` control software, pertaining the boards' trigger and data acquisition (TDAQ) systems. We analyse the gathered data across four diagnostic metrics, which reflect the general health and reliability of the HGCROC boards. We use these results to evaluate the tested boards, identifying common sources of read-out irregularities. Moreover, we use the results to examine the chip boards' control software and suggest potential improvements. Lastly, we aim to develop an automated anomalous-data identification algorithm for detailed HGCROC-board diagnostics.

1.3 Thesis outline

Section 2 provides a comprehensive overview of the history and evolution of particle DM searches. It is particularly focused on the thermal freeze-out hypothesis, which serves as the basis for LDM interactions with Standard Model particles.

Section 3 comprises a detailed description of the LDMX experimental design. This section delves into the inner-workings of the HCal and the role of the HGCROC.

Section 4 focuses on the HGCROC itself, laying the groundwork for understanding the chip's noteworthy electrical components responsible for its read-out.

Section 5 constitutes the main part of the thesis, comprising descriptions of the Lund HGCROC-board testing setup, the methodology and the results. The section contains both a general overview of the results, and discussion of potential sources of read-out irregularities.

Section 6 provides an outline of the automated irregularity identification algorithm, which was designed with a specific type of read-out anomaly found in the previous section.

Section 7 concludes the thesis with a general summary of the results and outlook into future research that may further improve HGCROC's reliability.

2 The Dark Matter Problem

2.1 Introduction

Though modern discussion of DM often involves quite revolutionary ideas, the initial guesses as to its nature were far more constrained. ‘Dark’ stars, planets and nebulae¹ marked the primary suspects for anomalous cosmological observations. Lord Kelvin estimated some 10^9 stars resting within our galaxy were far too dim to discern with existing instrumentation [1]. Studies of galaxy clusters and galaxy rotation curves only strengthened the hold of DM on cosmology. However, recently the discussion has shifted from massive celestial objects to tiny, invisible particles - the ‘dark matter’ proper-noun.

2.2 Particle Dark Matter

As discussed today, DM has a few notable traits:

- **Stability:** As observational evidence for DM can be found in the Cosmic Microwave Background and large-scale cosmological structures, DM must date back to at least the early stages of the Universe [2]. This implies DM possesses an extremely long lifetime.
- **Mass:** To exude gravitational influence, DM must be massive, meaning its mass is non-zero. Observations of dwarf galaxies, largely dominated by DM, have led to a generous mass (m_{DM}) range of

$$10^{-21}\text{eV} < m_{DM} < \sim 10^{23}\text{eV}.$$

We can derive the lower-limit by constraining the de Broglie wavelength of DM to fit a small dwarf galaxy. We find the upper-limit by considering the typical mass of such galaxy, roughly corresponding to 10^{37} kg [2].

- **(Near-)lack of SM interaction:** DM is defined by its lack of (or extremely feeble) interaction with matter through SM forces, particularly the electromagnetic force [2].

With these features serving as the foundation for DM candidates, we may retroactively dismiss some initial SM hypotheses. Neutrinos, for example, though electrically neutral and stable, could not have formed large-scale cosmological structures. This is due to their high energy during the early stages of the Universe [1]. By comparison, decoupling from SM candidates opens up the mass range considerably, as illustrated in Figure 1. Instead of tackling it all at once, we may select a specific sliver by adopting one of the many DM-frameworks, which impose some basic assumptions. One such framework is the thermal freeze-out hypothesis.

¹Later known as Massive Astrophysical Compact Halo Objects (MACHOs) [2].

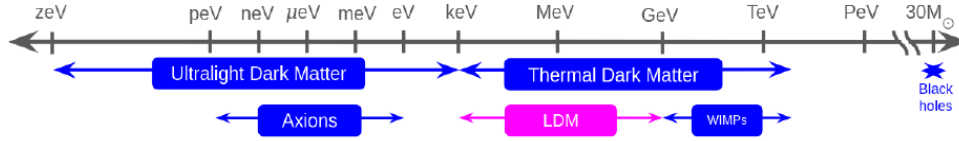


Figure 1: Energy ranges of possible DM candidates. Notably, the Thermal DM region is divided into LDM (keV-GeV) and WIMPs (GeV-TeV). Figure sourced from [3].

2.3 Thermal freeze-out

To explain the relative abundance (and longevity) of DM, the thermal freeze-out hypothesis posits that DM is a ‘thermal relic’ from the early Universe. In this paradigm, both SM and DM particles were kept in chemical and thermal equilibrium. As the Universe cooled down, however, the number of DM particles started diminishing through some annihilation process. The annihilation rate slowed down as the expansion of the Universe separated the remaining DM [4][5]. At the same time, particles with more modes of interaction kept annihilating, leading to a substantial gap between the abundance of SM and DM particles. Thus, we can determine the rough DM-annihilation cross-section, and hence the possible energy regime of its mediator. Though the actual range is extensive, the possibility of DM interacting via the Weak force quickly established itself as the initial favourite. Thus, the search for Weakly Interacting Massive Particles (WIMPs) was born.

With WIMPs, one possible channel is $2\text{SM} \rightarrow 2\text{DM}$ self-annihilation (see Figure 2). The WIMPs annihilate, releasing a Weak-force mediator, $W^\pm$ or Z^0 , which then decays into two SM particles. The WIMP hypothesis is perfect in its simplicity, requiring only one new particle to be added with no strings attached. Super-symmetric extensions of the SM also predicted particles that closely matched WIMPs’ description - a coincidence that became known as the ‘WIMP miracle’. However, it was too good to last. Decades of research have only managed to constrain the WIMP mass-range, with the elusive particle yet to manifest [6]. Still, the WIMP hypothesis only covered a portion of what is now called thermal dark matter. With mounting evidence against the WIMP hypothesis, and now-improved sensitivity of particle detectors, we can explore deeper into the lower thermal regime.

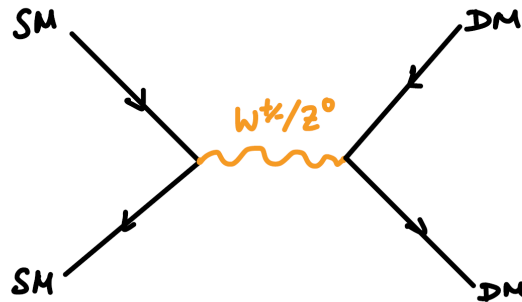


Figure 2: Feynman diagram sketch of a $2\text{SM} \rightarrow 2\text{DM}$ interaction, mediated by $W^\pm, Z^0$ bosons.

2.4 Light Dark Matter

Moving past WIMPs, we enter the Light Dark Matter² energy range. Removing the mass constraints on the LDM-SM mediator allows LDM to act through a new force. As seen in Figure 3, the general mechanism for LDM-SM interaction is similar to that of WIMPs, with one major difference. Instead of the $W^\pm$ or Z^0 bosons, the interaction is mediated via a new A' gauge boson - the ‘dark photon’ [7]. LDMX is designed to capture a version of this process, called ‘dark bremsstrahlung’. In Figure 4, we envision an electron scattering off of a nucleus. Similarly to a standard bremsstrahlung process, the electron emits radiation, losing a sizeable portion of its momentum. In this case, the emitted particle is a dark photon, which decays into two DM particles. As both the mediator and DM are invisible, the process appears to an outside observer as the electron losing its momentum to seemingly nothing. Therefore, we can *infer* the presence of dark matter using conservation of momentum.

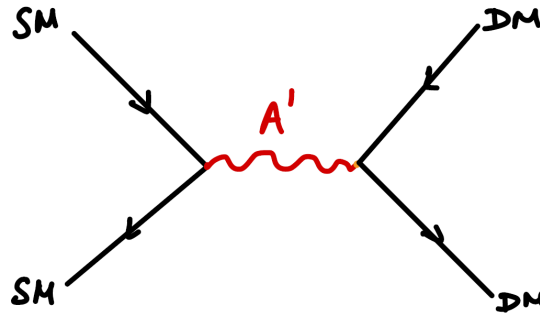


Figure 3: Feynman diagram sketch of a $2SM \rightarrow 2DM$ interaction, mediated by a dark photon (A').

Naturally, dark bremsstrahlung must be incredibly rare (as we have yet to observe it), but it is also *possible* and *measurable* - what else does an experiment really need?

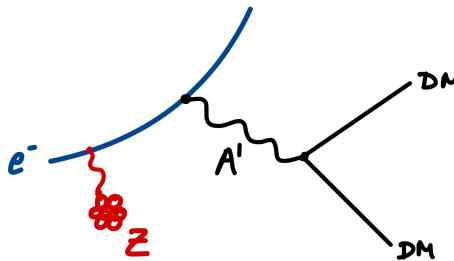


Figure 4: Sketch of a hypothetical SM-DM interaction through a dark photon (A') released by a scattered electron (e^-). The emitted particle, A' , is converted into two DM particles, which do not directly interact with SM forces, and hence are ‘invisible’. Figure inspired by [3].

²Sometimes also called Feebly Interacting Massive Particles, or FIMPs.

3 The Light Dark Matter eXperiment

3.1 Introduction

Following the WIMP-dominated era of the search for DM, LDMX is designed to probe the sub-GeV energy range [5]. The experiment will be largely assembled with repurposed instrumentation from existing experiments (like CMS, Mu2e, etc.), as seen in Figure 5. Within only a few years of continuous operation, LDMX will greatly expand current LDM-mass experimental limits (see Figure 6) [5]. To achieve this, the experiment comprises a multitude of trackers and calorimeters, monitoring an 8 GeV electron-beam scattered off of a fixed thin tungsten target, searching for dark bremsstrahlung events [5]. As described in Section 2.4, though the actual dark photons will be undetectable, the dark bremsstrahlung process can be identified by its unique fingerprint - inexplicable missing momentum of the scattered electron. To that end, the various detectors are designed to track and measure the energies of incoming and recoil electrons. At the same time, they are used to veto, or exclude, any events that do not match the dark bremsstrahlung process - a task easier said than done.

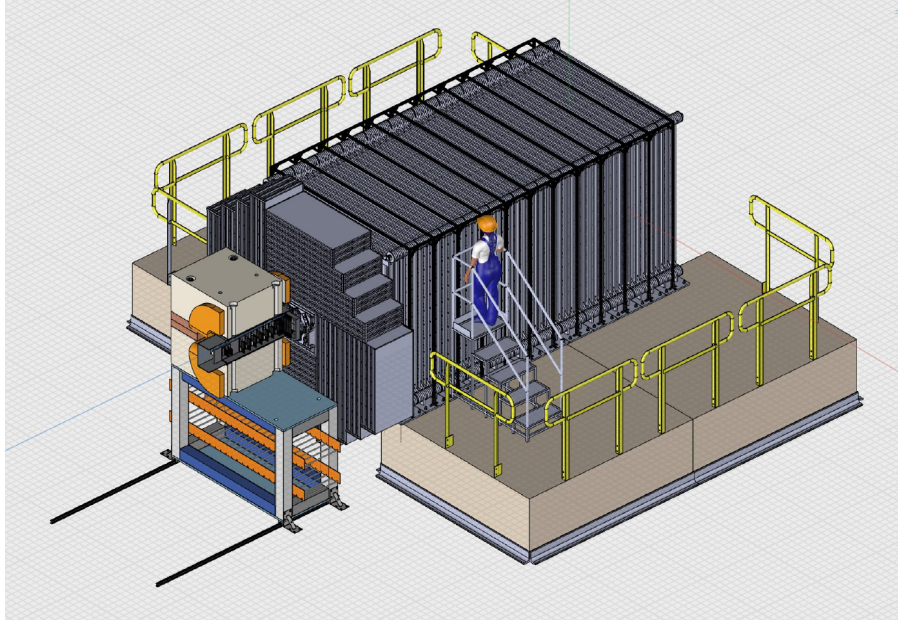


Figure 5: LDMX detector 3D-model with human to scale. From left to right, the model shows the trigger-scintillator and tracker systems, leading into the target and ECal, surrounded and followed by HCal layers. Figure sourced from [5].

3.2 Experimental design

To fully understand the LDMX detector design, we can follow the path of the electron-beam. The electrons, sourced by the Linac-to-End-Station-A [8], are accelerated within the LCLS-II facility into the thin tungsten target. As they reach LDMX, they first enter a large spectrometer magnet, illustrated in Figure 7. The magnet surrounds the trigger-scintillator and the target sandwiched between tagging and recoil trackers. These subsystems monitor the number, trajectory and momenta of the electrons, before and after scattering. The magnet produces a uniform field around the tagging trackers to filter out low-energy impurities in the beam. This field diminishes in strength around the recoil tracker to accept low-momentum recoil electrons, often deflected at wide angles [5].

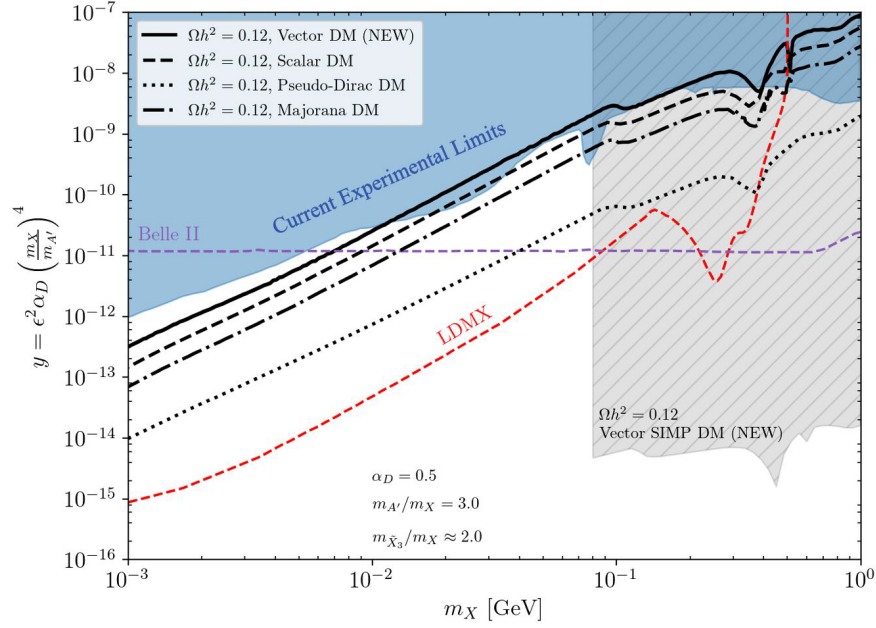


Figure 6: Current experimental limits on the DM interaction scale (y) as a function of DM-mass (m_X). The LDMX results are projected to expand these limits (marked in blue) up until the red-coloured curve. Figure sourced from [5].

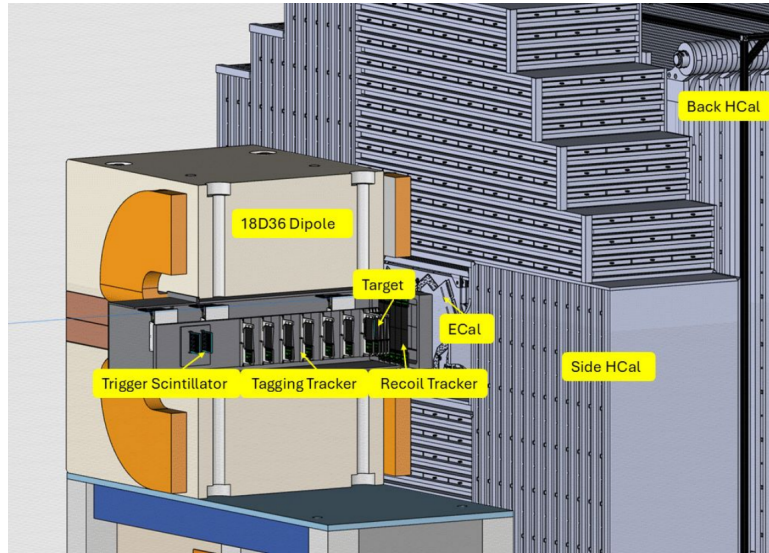


Figure 7: Close-up of the front section of the LDMX detector. The dipole magnet encompasses the trigger-scintillator, the tagging trackers and recoil trackers. The target is positioned between the trackers, and marks the beginning of the ECal. The ECal is surrounded and followed by the HCal. Figure sourced from [5].

The front section is immediately followed by the ECal and the HCal. The ECal measures the final energy of electrons and photons. In contrast, the HCal mainly focuses on hadrons, unaccounted for by the ECal (see Figure 8).

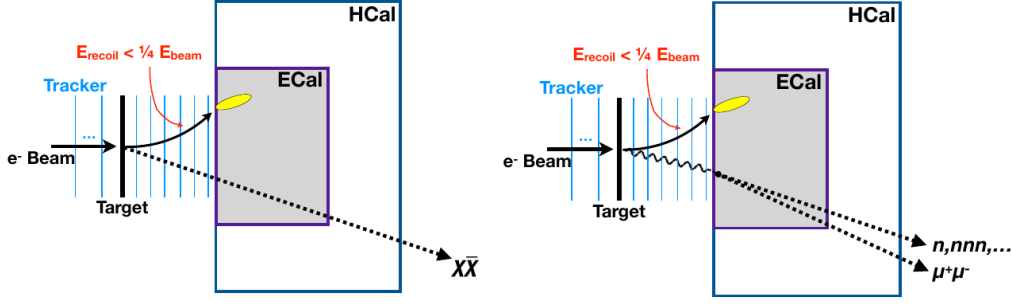


Figure 8: Illustration of different electron-scattering results. On the left, the figure presents a sketch of LDM production ($X\bar{X}$). On the right, the figure shows a possible photo-nuclear interaction, producing neutrons or muon-pairs. Figure sourced from [5].

3.2.1 Expected signals & HCal design

Despite the experiment deliberately setting-up favourable conditions for electron scattering, standard (not to mention dark) bremsstrahlung events are still fairly rare. Indeed, most of the high-energy electrons will not undergo scattering at all and will instead simply shoot through the thin tungsten target [5]. Other background sources range from numerous particle showers to rare photo-nuclear effects, as seen in Figure 9.

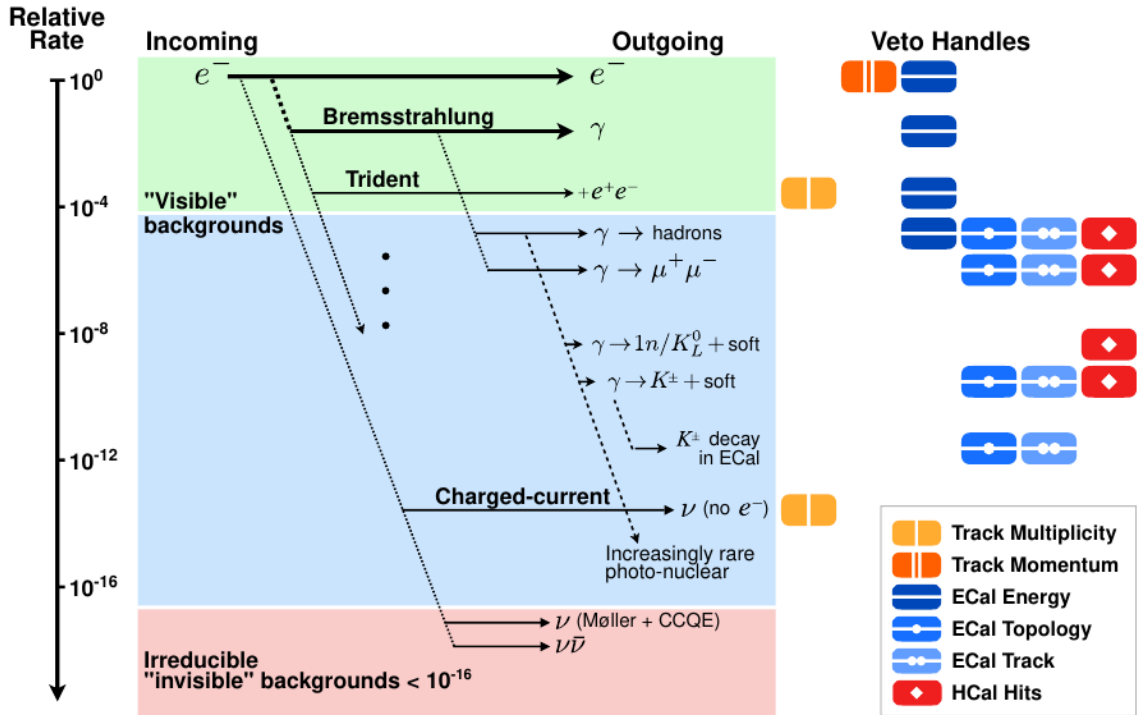


Figure 9: Relative rate of notable background signals in LDMX. The signals are split into “visible” and “invisible” to LDMX detectors, with neutrinos (ν) comprising most of the invisible backgrounds. We note the signals relevant to the HCal (marked with diamond shapes on red backgrounds) can vastly vary in relative rates. Figure sourced from [9]

Photo-nuclear reactions may take place both within the target and ECal, producing heavy neutral particles or muons. Electrons can also undergo wide-angle bremsstrahlung, which necessitates a specific detector geometry. Even if no rare interactions take place within the detector, the capricious cosmic rays may hit a component at any time.

As most of these events will escape the ECal, the HCal plays a crucial part in ensuring LDMX's high veto-power. Seen in Figure 5, the HCal comprises two major parts: the Side HCal and Back HCal. The former is coiled around the ECal in a pin-wheel fashion, while the latter extends behind it [5]. Both parts operate on the same principle - steel absorber plates 'catch' energetic particles, resulting in a shower of particles. These are then detected by plastic scintillators. For the Back HCal, this translates to twelve eight-layer modules of steel plates. Ten 'quad-counters' - units of four scintillator bars - are glued to each layer. The layers alternate between X and Y orientations [5]. Similarly, the Side HCal comprises four modules, each with 24 layers of steel absorbers and quad-counters arranged in X-Y and Y-Z orientations. The Back HCal serves as the main high-efficiency veto system. It is primed to detect highly energetic neutral particles, cosmic ray muons, and a small fraction of charged particles not rejected by the ECal. Though the Back HCal covers most of the calorimeter, the Side HCal is necessary to catch wide-angle bremsstrahlung radiation, which could avoid the ECal altogether.

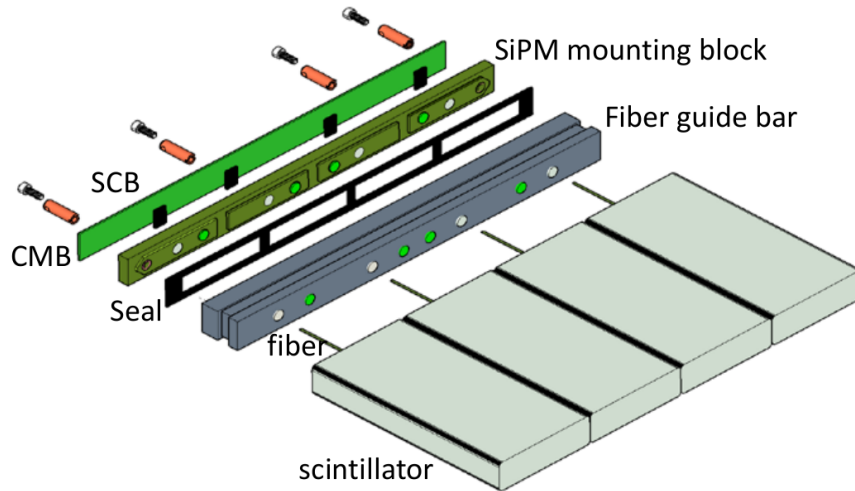


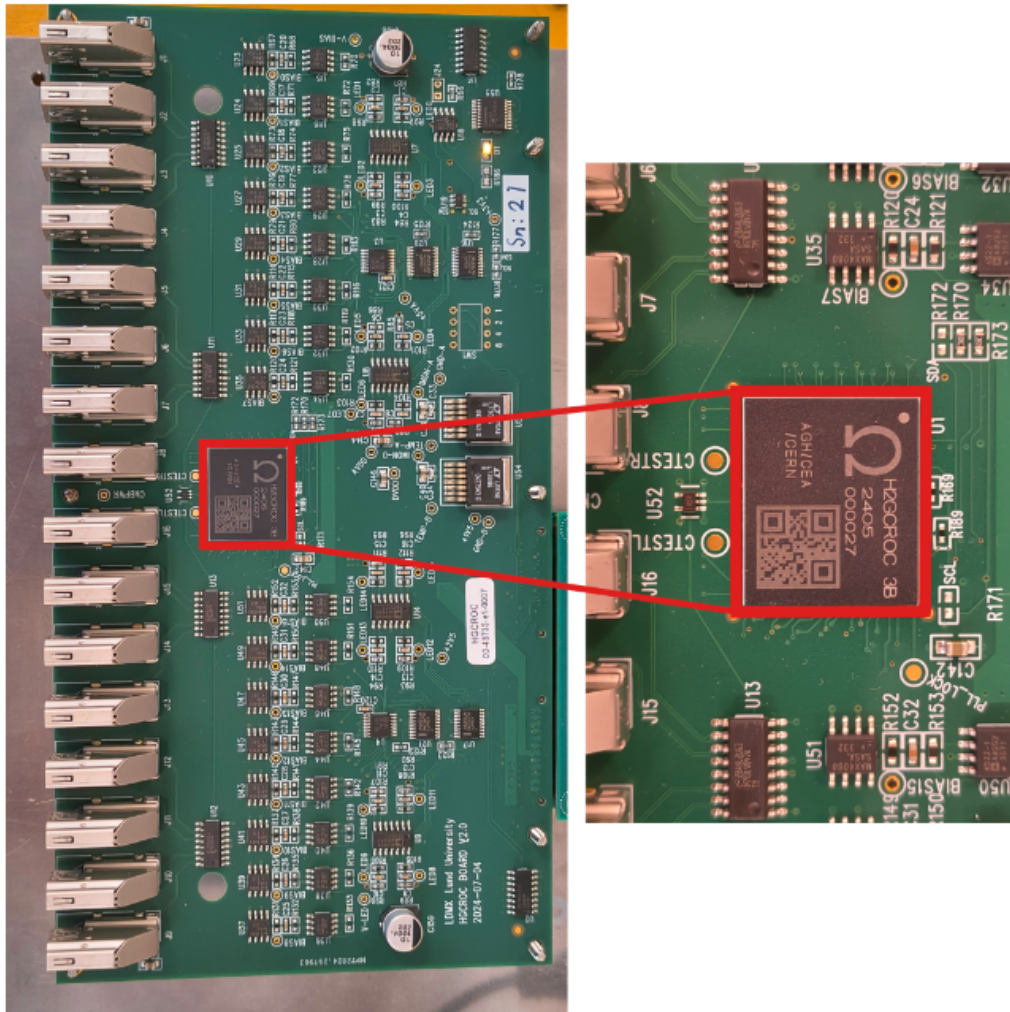
Figure 10: Model of the edge of a quad-counter connected to a CMB and SiPM via wavelength-shifting fibre. The quad-counter comprises four scintillator bars, each ran-through with fibre. The SiPMs are mounted on SiPM carrier boards (SCBs). Figure sourced from [10]

After a particle passes a quad-counter, the light from the scintillator bars is directed through wavelength-shifting fibre to Si photo-multipliers (SiPMs). The SiPMs are mounted in groups of four on Counter Mother Boards (CMBs), which are located at the end of each bar (see Figure 10) [5]. The output SiPM signal is then transmitted into an HGCROC, which serves as the initial signal read-out system. The HGCROC boards housing the chips are mounted on 'backplane' boards. Mezzanine boards are also mounted on the backplane. They interface to the external TDAQ system of LDMX. The backplane enables communication from the HGCROC board all the way to LDMX's central TDAQ and controls. HGCROC failures or component-to-component variations can cause cascading effects on the HCal's overall veto power, impacting LDMX's sensitivity. Thus, this study focuses on characterizing the performance and consistency of the HGCROC boards.

4 H2GCROCV3: Design and Function

4.1 Introduction

As a particle hits one of the many scintillator bars of the HCal, scintillated light is captured and routed via fibres into a SiPM. In turn, the SiPM produces an electric signal that is sent into one of the many channels of an HGCROC, seen mounted on an HGCROC board in Figure 11. The chips, introduced in Section 3.2.1 are adapted from the CMS experiment [5]. Specifically, they are chosen from the version-3 production class. This study concerns the H2GCROCV3b ASIC.



37 MHz), reading the signals at 40 MHz. Upon entering the chip, the SiPM signal is attenuated by a current conveyor (CC), after which it undergoes charge-to-voltage conversion and initial shaping by a pre-amplifier (PA) and shaper circuit³. Finally, the signal is digitized and stored in a buffer, which later passes it on to an external TDAQ system. This study focuses on the behaviour of the DAQ path, pictured in the green box in Figure 12.

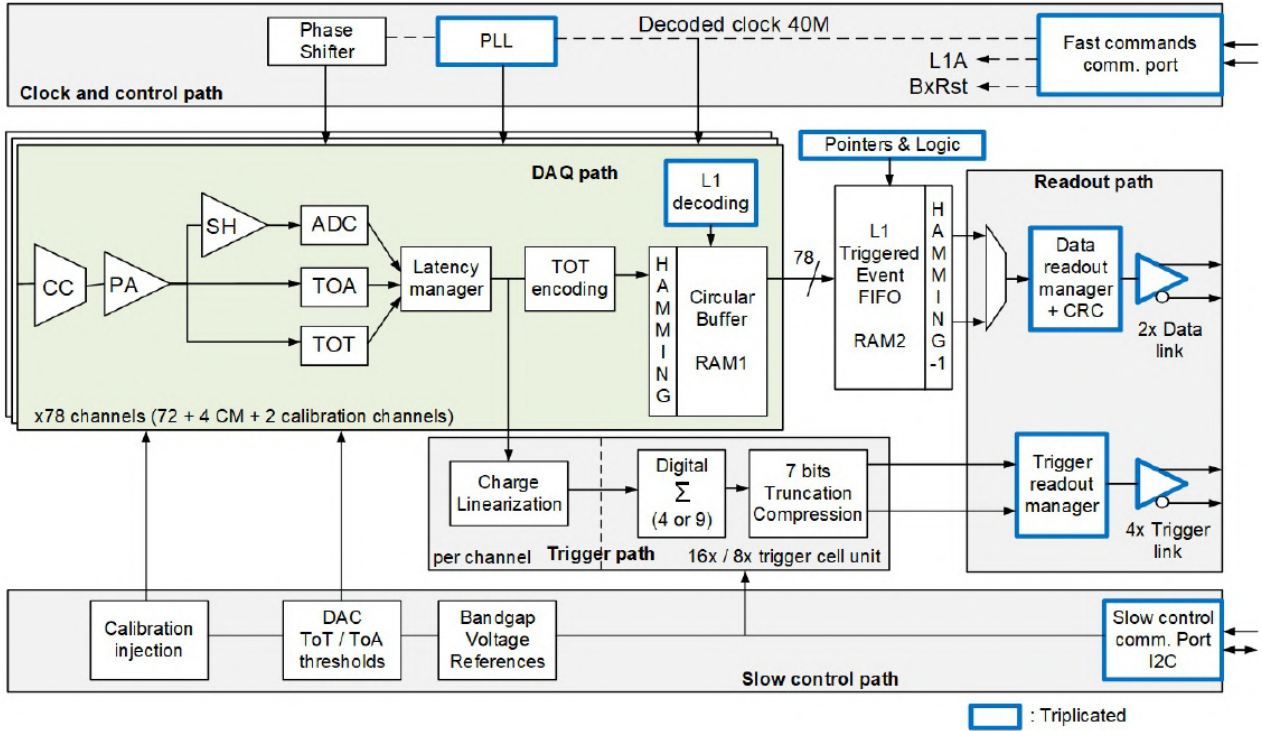


Figure 12: Simplified schematic of the HGCROC. Each box corresponds to a different “path” of the ASIC. The paths contain components responsible for the chip’s various features. Notably, the DAQ path components perform the initial signal shaping and digitization, storing the results in a buffer. The buffer is connected to the readout path, which sends the signal to external electronics. Figure sourced from [11].

4.3 Digital converters

The DAQ path comprises three digital converters - one analogue-to-digital converter (ADC) and two time-to-digital converters (TDC). The ADC digitizes the pulse magnitude over 10 bits, saturating at 1023 ADC units. The TDCs measure the time-of-arrival (TOA) and time-over-threshold (TOT). The 10-bit TOA-TDC marks the moment at which the pulse *arrives*, that is, when its magnitude sufficiently exceeds the baseline signal, called the *pedestal*⁴. The 12-bit TOT-TDC, on the other hand, calculates the time it takes for the pulse to fall below some arbitrary threshold after triggering TOA, effectively estimating the pulse-length.

³To elaborate, the shaper circuit is there to both further shape the signal, as well as ensure the signal-to-noise ratio stays below 20% after 25 ns, such that subsequent signals do not overlap.

⁴The ADC, like many other components, is deliberately biased with some voltage. This results in a non-zero ADC-signal even when no pulse is present, marking the ADC-read-out baseline.

4.3.1 Analogue-to-digital converter

The ADC operates at 40 MHz, splitting the pulse into 25 ns sections referred to as “bunch-crossings”⁵ (bxs), as seen in Figure 13. The bxs are then split into up to four, eight or 16 phases. The number of phases can be configured manually [11]. The phases differ in length based on this setting. In this study, the bx counts 16 phases, each 1.56 ns long. As the ADC can only output one value at a time, it is configured to choose a specific phase. Using ‘multi-sampling’, the ADC digitizes one phase of each bx (all selected phases in Figure 13). This digitization is done with a single pulse. In contrast, ‘single-sampling’ considers only one bx at a time, requiring additional pulses to be sent for each subsequent bx (only one selected phase in Figure 13 per pulse). The continuous signal is thus reduced into only a handful of values, which are stored in the circular buffer. Finally, if the ADC data passes some threshold, a positive level-1-acceptance (L1A) signal is sent to release it to the external TDAQ system.

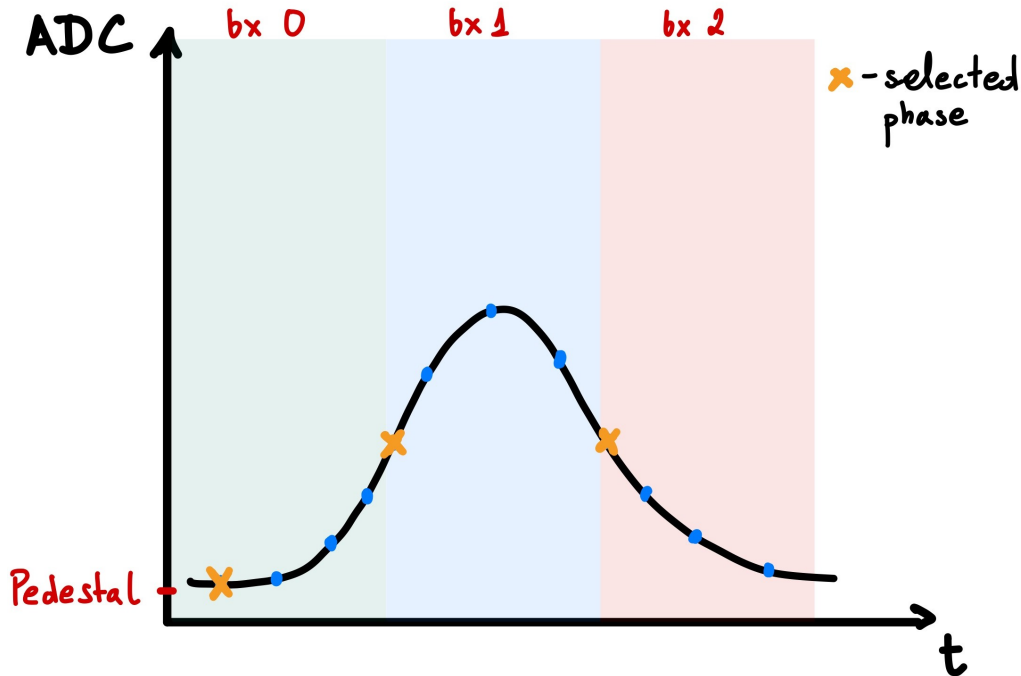


Figure 13: Sketch of pulse digitized in over three bxs. In this example, each bx contains four phases, highlighted as blue dots on the pulse. One phase (marked with ‘x’) is selected per bx during the analogue-digital conversion process to be read out by the ADC. Both the pulse magnitude and bx widths are not to scale.

4.3.2 Time-to-digital converters

The TDC is split into two converters, both of which digitize time passed, rather than voltage. The TOA-TDC marks the arrival of a pulse, while the TOT-TDC calculates the length of the pulse (see Figure 14). Notably, the TOA and TOT enable the ADC to measure past its saturation-point. As the TOT is mostly linear after ADC-saturation, it can be used to extrapolate ADC values well past the ADC’s initial limits. This enables the HGCROC to accurately measure the peaks of very large pulses, ensuring the HCal’s high veto-power.

⁵Though no bunches actually cross in the LDMX, the term was inherited from LHC experiments.

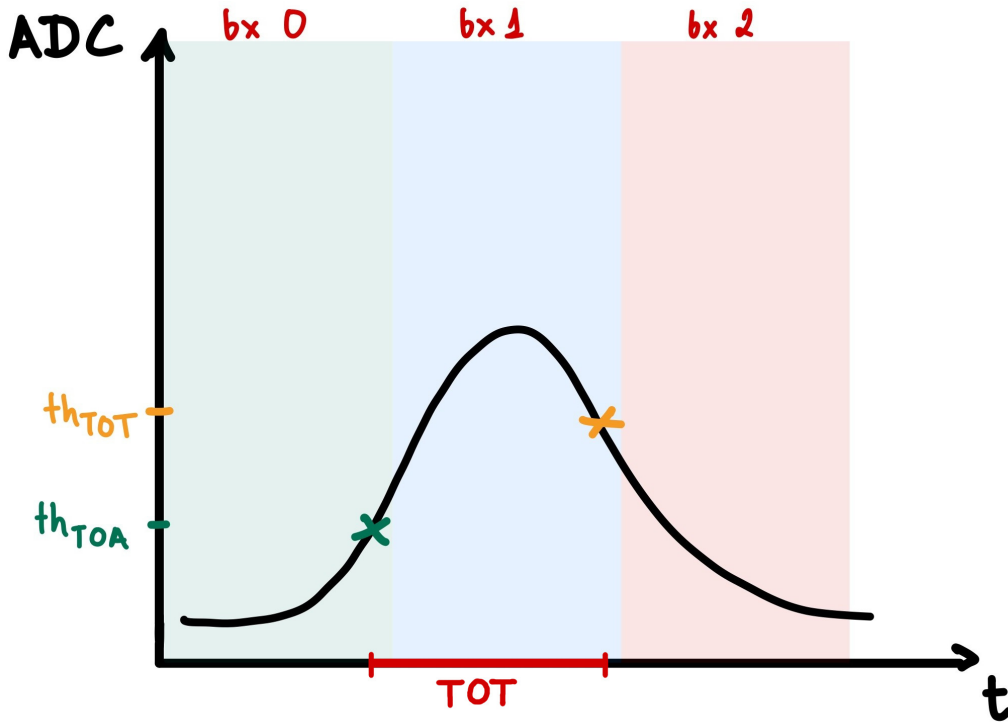


Figure 14: Sketch of a charge-injection pulse with TOA and TOT thresholds (th_{TOA} , th_{TOT}). The TOT (effectively the pulse-length) is defined as the time between the two phases marked with ‘x’ on the figure.

4.4 Internal charge-injections

The HGCROC contains its own internal charge-injection circuit for calibration and testing. The circuit, seen below in Figure 15, can be split into two digital-to-analogue converters (DACs). The DACs inject charge corresponding to a 12-bit CALIB scale. Choosing one of the DACs, the user is able to select a high-range (HR), low-range (LR), or pre-CC mode. As the name suggests, the pre-CC mode corresponds to the 3 pF capacitor set before the CC. As the CC attenuates the pulse to a fraction of its original magnitude, the pre-CC signal is the weakest of the three. The LR and HR modes, on the other hand, engage the remaining 500 fF and 8 pF capacitors. The HR mode can also engage the pre-CC capacitor at the same time. Though the LR signal is similar in magnitude that of the pre-CC mode, the two can be compared to evaluate the CC itself. This study mainly focuses mainly on the HR and pre-CC outputs.

4.5 Pflib: control software

The Pretty Fine Library (pflib) is the custom read-out software enabling communication between a computer and the HGCROC. For the purposes of this study, we can focus on a specific part of the library called `pftool`, which mainly serves as the software interface. `pftool` comprises a wide variety of functions, ranging from calibration to data acquisition, manually chosen by the user. Users can start `pftool` with a pre-written file to automatically run a set of functions.

Although the interface is indeed pretty fine, some settings of the chip cannot be directly adjusted via `pftool`. Instead, a `yaml` configuration file has to be manually written, and subsequently applied to the HGCROC via the interface. The settings contained within these files range from turning specific

components of the chip on or off, setting reference voltages for said components, or even delaying the read-out of the ADC. This is important, as facilities around the world will often use different configurations of the chip, depending on their experimental setups.

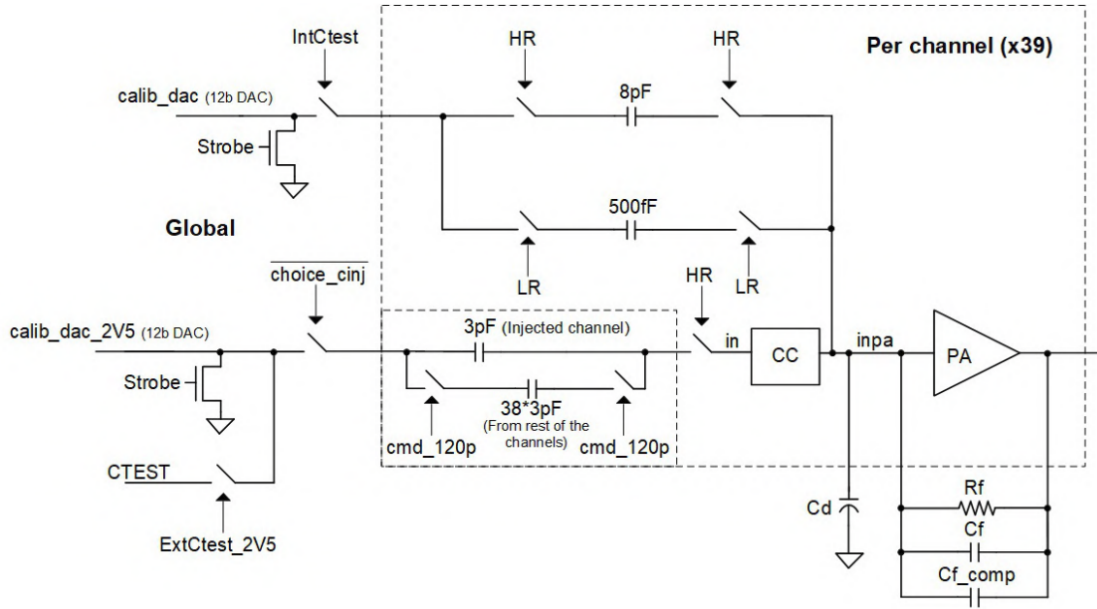


Figure 15: Schematic of the internal charge-injection circuit, comprising two DACs. The CALIB2V5 DAC is responsible for pre-CC charge injections using the 3 pF capacitor. The CALIB DAC is responsible for HR and LR injections, using 8 pF and 500 fF capacitors, respectively. The injected pulse is directed to a PA, after which it is read out by the ADC and TDCs. Figure sourced from [11]

5 H2GCROCV3b: Characterization

5.1 Introduction

The HGCROC is a small, complex ASIC. Its reliability is an important part of the HCal veto-power. Minor differences between chip components or initial `pflib` settings can drastically affect its the read-out. To ensure standardization and proper functionality during the experiment, the chip must be tested in various conditions. This study concerns preparing a new, generalized testing procedure to evaluate the internal charge-injection response of 10 different HGCROC boards, each featuring a unique H2GCROCV3b ASIC. The results of this investigation focus on four metrics, which characterize the **ADC**, **TOA-TDC**, and **TOT-TDC** components of the chips.

5.2 Testing setup

The Lund HGCROC testing setup (see Figure 16) differs from the final LDMX configuration, as described in Section 3.2.1. The HGCROC board is connected to an FPGA development board - the ZCU102 - via a custom adapter, bypassing the backplane. The ZCU102 directly controls parts of the TDAQ via its firmware. Notably, the firmware version used in this study does not allow for multi-sampling. Instead, we carry out the testing via internal single-sampled charge-injections. This results in the ADC outputting one phase of one bx per pulse. To ensure full-coverage of all 16 phases of each bx, we use a large number of identical pulses sent every 25 ns. Though not ideal for recreating experimental conditions, this method maximizes the likelihood of catching ADC and TDC read-out irregularities. Furthermore, the lack of a backplane allows us to test the HGCROC board directly, providing an independent source for comparison with other LDMX testing sites.

5.3 Methodology

The chip boards are referred to by their chip-board numbers, ranging from 20 to 30. Each one was tested using a series of ‘scans’, chosen via the `pftool` interface. The scans, denoted as a different `pftool` ‘tasks’, correspond to specific charge-injection modes and read-out configurations. As an example, the `PARAMETER.TIMESCAN` task induces a charge-injection within a chosen channel of the HGCROC. The injection can be set to a given mode (HR, LR, pre-CC) and magnitude, set by a specific `CALIB` value. A sequence of such scans was coalesced into a single testing procedure, focused on four evaluation metrics.

5.3.1 Evaluation metrics

Fundamentally, the read-out of the HGCROC depends on the ADC, TOA-TDC and TOT-TDC. Though measuring different signal parameters, they all serve to digitize the peak-magnitude of an incoming pulse. The ADC converts the signal up to 10-bits, and the TDCs help it overcome that boundary. The testing procedure collects data evaluated on four metrics to assess the ‘health’ of the ASIC, focusing on the aforementioned components. The data is collected with multiple samples per channel, i.e. the scans are performed multiple times. As such, we note a distinction between *channel-wise* and *link-wise* metrics. We refer to *channel-wise* values when considering the distribution of samples within a single channel. We refer to *link-wise* metrics when looking at the distribution of sample-means across channels 0-35 (link 0) and channels 36-71 (link 1).

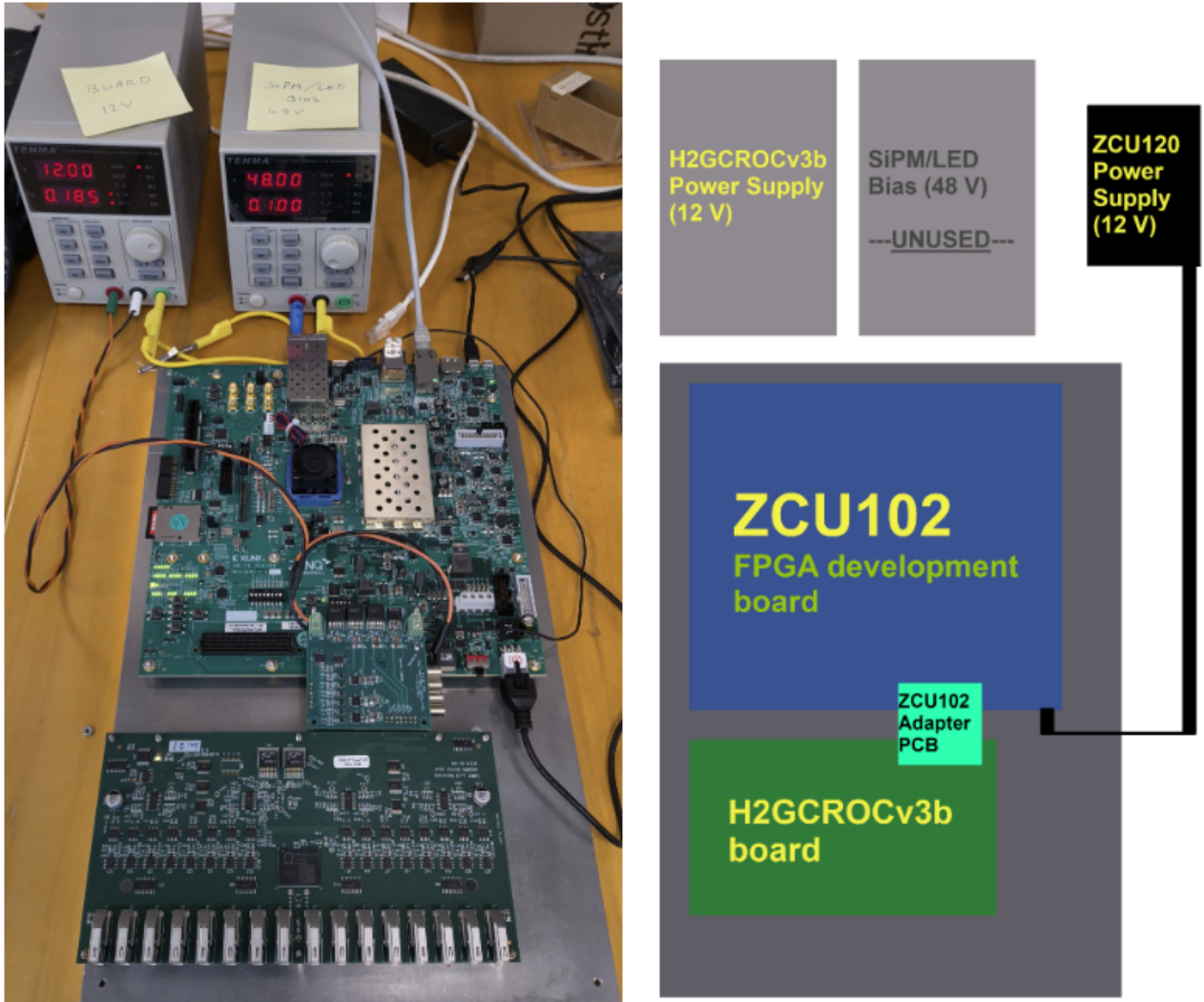


Figure 16: The Lund HGCROC testing setup, pictured on the left and simplified on the right. The H2GCROCV3b board is connected to the ZCU102 via a custom adapter PCB. The ZCU102 is connected to the internet, and hence `pflib`, with an ethernet cable. Both the H2GCROCV3b board and the ZCU102 are powered via external 12 V power supplies. Notably, power supply providing the 48 V SiPM/LED bias is not connected, as it is unnecessary for internal charge-injection tests.

- **Pedestal alignment:** Ideally, the ADC-pedestals should be equal within each link. We call the pedestals “aligned” when their link-wise root-mean-square deviation (RMSD)⁶ does not exceed 10 ADC units. We further examine the channel-wise RMSD using the same threshold when evaluating the response of specific channels.
- **Peak-ADC alignment:** Similarly to pedestals, the ideal behaviour of ADCs across each link should result in the same peak-ADC (meaning peak-ADC value), for the same amount of injected charge. We evaluate peak-ADC alignment by considering their channel-wise and link-wise RMSD values.
- **ADC/TOT linearity:** As described in Section 4.3.2, both the ADC and TOT values should follow a positive linear trend over a CALIB range. Ideally, the TOT values begin and follow

⁶For derivation, see Appendix A.

that relationship right after the ADC saturates. In reality, the TOT initially follows a slowly-increasing curve before entering its ‘linear regime’. Hence, we evaluate the ADC/TOT linearity by using linear regression in both converters’ linear regimes, with the metric being the R^2 -value of the fits (see Appendix B).

- **TOA response:** Contrary to the other metrics, the TOA response over a range of CALIB values is difficult to consider quantitatively. Instead, we visually check the TOA evolution (‘time-walk’) using aggregate plots, which show time-walks of all channels within each link simultaneously.

5.3.2 Testing procedure

In order to take the data necessary for properly evaluating the H2GCROCV3b read-out, we used a standardized testing procedure composed of specific `pftool` tasks. We can split the procedure into two parts: configuration and data acquisition. Starting with configuration, we use a sequence of `pftool` functions to:

1. **Level the pedestals**, by aligning each channel’s pedestal with respect to a calibration channel on each link. This function runs before and after lowering the pedestals;
2. **Lower the pedestals**, by setting specific reference voltages within the shaper-circuit on each channel. This allows for a wider ADC-range while minimizing the influence of noise-fluctuations;
3. **Configure the TDCs**, by setting unique TOA and TOT reference voltages, called ‘thresholds’, on each channel.

The data-acquisition procedure comprises:

1. **Pedestal scans**, taken after both instances of alignment, with 100 samples per channel;
2. **ADC time-scans**, taken after configuration with 50 samples⁷ per channel over five bxs. The scans involve HR and pre-CC charge-injections over the following CALIB values:
 - {64, 128, 256, 512} for HR,
 - {256, 512, 1024, 2048, 4095} for pre-CC.

Though it is possible to reach higher CALIB values for HR scans, we limit the range to 512, as the pulses tend to saturate the ADC after $\text{CALIB} \sim 450$.

3. **ADC/TOA/TOT CALIB-scans**, which engage the ADC and the TDCs at the same time using HR charge-injections. The scans take data over two CALIB ranges, 0-1080 (steps of 20) and 1080-4080 (steps of 100), and cover 10 samples per channel over two bxs.

Each of these functions was used in a standardized order, with the exact tasks and settings listed in Appendix C.

⁷Exceptionally, board 22 was scanned with a 100 samples.

5.4 Data analysis

Using newly developed analysis Python scripts from `pflib` (see Appendix D), we review the data taken via the aforementioned testing procedure. In particular, we focus on the diagnostic metrics listed in Section 5.3.1. As certain boards feature multiple irregularities across different metrics, the final results are summarized in Table 4 at the end of this section.

5.4.1 Pedestal alignment

Board/RMSD [ADC units]	Link 0 (BL)	Link 1 (BL)	Link 0 (AL)	Link 1 (AL)
Board 20	1.51	2.04	1.67	1.93
Board 21	1.76	1.87	2.02	1.46
Board 22	1.69	1.17	1.58	1.18
Board 23	2.10	1.51	1.81	1.86
Board 24	1.81	1.57	1.67	1.20
Board 25	26.9	2.59	26.7	2.23
Board 26	2.23	4.64	1.74	4.91
Board 27	1.79	2.11	1.74	1.40
Board 28	2.35	1.89	3.18	1.78
Board 29	1.43	2.88	1.99	2.45
Board 30	2.51	1.76	1.91	2.00

Table 1: The link-wise RMSD of mean-pedestals over each link for HGCROC-boards 20-30. The results are split into before-lowering (BL) and after-lowering (AL) of pedestals. The pedestal means are taken for a 100 samples on each channel. The RMSD is calculated in ADC units and rounded to 3 s.f.. Results over 10 ADC units are marked with red.

For pedestal-alignment evaluation, we consider pedestal scans taken before and after lowering. As shown in Table 1, the overwhelming majority of the pedestals, scanned on HGCROC boards 20-30, seem to be properly aligned, with their RMSD values per link falling well below the 10 ADC-unit threshold. Looking at Figure 17 from HGCROC board 22, for example, we can see the mean-pedestals align fairly well across link 0, with the channel-wise RMSD below 2 ADC units.

Channel misalignment Misaligned channels mark the most noticeable type of pedestal-alignment anomaly, visible on boards 25, 26, 28 and 29. Board 25 stands out as the most drastic example, with a suspiciously high RMSD over link 0, highlighted in Table 1. Looking at Figure 17, we can see a misalignment between channel 19 and the rest of the link. As the RMSD of the channel 19 sample distribution is well below the threshold, we may rule out the possibility of an outlier within the channel 19 samples bringing down the channel’s pedestal mean. Furthermore, the irregularity appears before and after the lowering procedure. Using this information, we can reasonably conclude that the high link 0 RMSD of board 25 is due to a malfunctioning channel.

Although link-wise RMSD at a 10 ADC-unit threshold was not sufficient for catching all pedestal irregularities in the tested boards (see Appendix F), we believe it is a useful diagnostics tool. Further work could include optimising the threshold level.

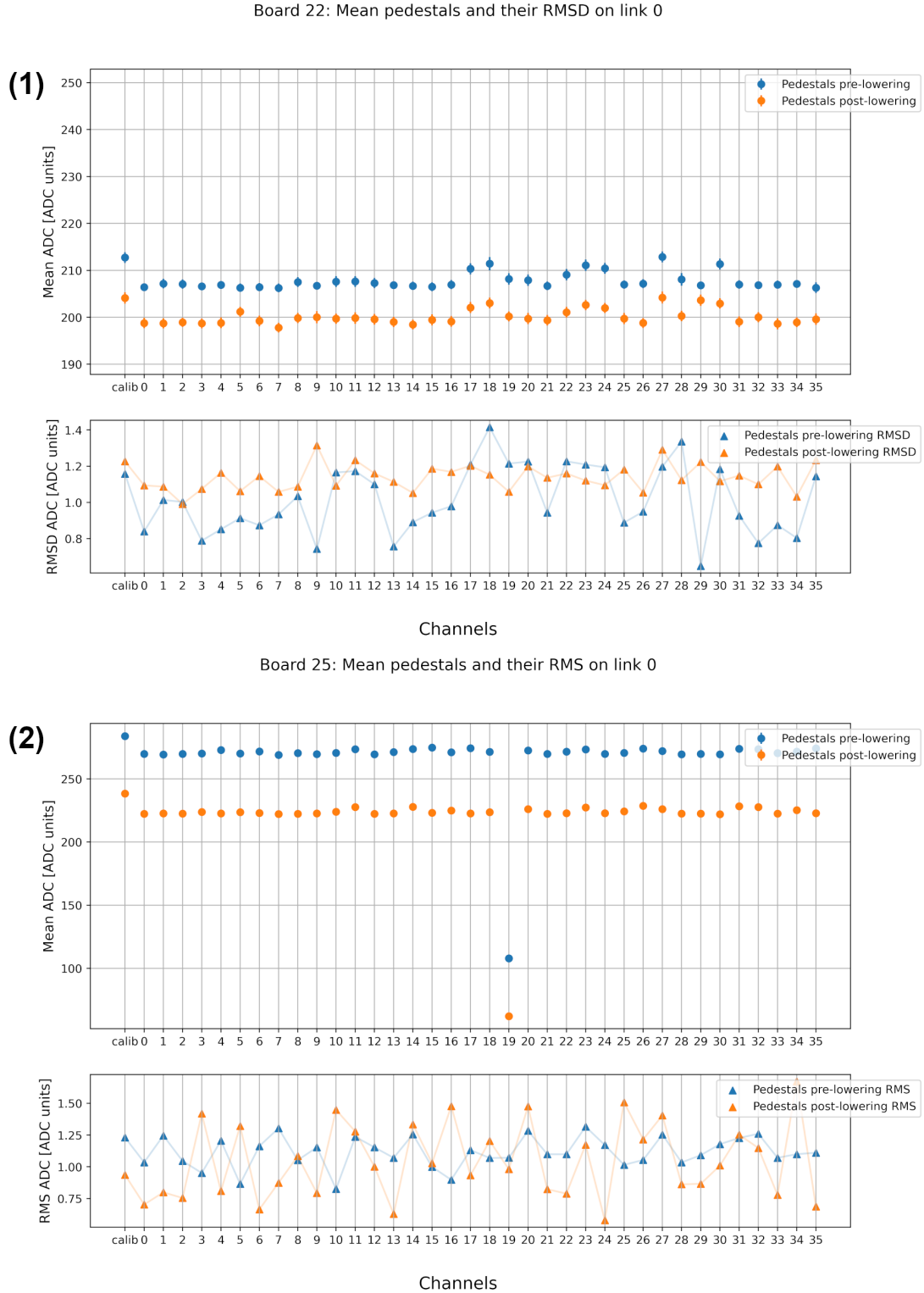


Figure 17: Plots of mean pedestals across link 0 on HGCROC boards 22 **(1)** and 25 **(2)**, before and after lowering. The means are taken over 100 samples on each channel. The plots are split into mean pedestal ADC values and their channel-wise RMSD values. **(1)**: A similar trend can be seen on link 1 in Appendix E.1. **(2)**: We note the misalignment between channel 19 and the rest of the link.

5.4.2 Peak-ADC alignment

To evaluate peak-ADC alignment, we track the link-wise RMSD of peak-ADC values from HR and pre-CC time-scans. As can be seen in Tables 2 and 3, overall, the HR scans seem to produce a healthier response than pre-CC scans. As an example, the HR results of HGCROC board 21 (see Figure 19) show well-aligned peak-ADC values across the CALIB range. Notably, CALIB 512 is excluded from the plots, as the corresponding ADC-peak values across all samples and channels are at the saturation point. Taking a closer look at the pulse-shape on one board-21 channel (see Figure 18), we see a relatively smooth curve traced by tight-knit sample distributions.

We observe a variety of symptoms across all the tested boards, with some subtle enough to avoid detection with link-wise RMSD. As such, we split the anomalous time-scan responses into multiple categories based on their link-wise RMSD and visual assessment.

Pedestal misalignment As shown in Section 5.4.1, some of the boards contain misaligned channels, with pedestal values below the rest of their link. This directly translates to peak-ADC misalignment on the same boards and channels. Taking board 25 as the most dramatic example (see Figure 19), channel 19 clearly lags behind the rest of link 0, even after the rest of the channels saturate at CALIB 512. As a result, board 25's link-0 RMSD results in Table 2 are abnormally high, while its link-1 RMSD remains well below the 10 ADC-unit threshold.

TOA/TOT triggers We found that active TOA/TOT-triggers during ADC operation result in the non-zero link-wise RMSD on boards 22 and 23, found at CALIB 512 in Table 2. Furthermore, as the TDCs trigger only at ADC values close to saturation, we can explain almost all of the abnormally-high RMSD results for pre-CC CALIB 4095, found in Table 3. During testing, we realized turning off both TDCs is crucial for obtaining accurate time-scan results for high CALIB values (see Appendix G). Unfortunately, we were not able to redo the pre-CC scans due to time constraints. Boards 22-24 were also distributed to different facilities.

Pedestal increase As Table 3 shows, HGCROC boards 21 and 22 account for some of the highest link-wise RMSD values from pre-CC time-scans. Looking at the response on board 21 in Figure 20, we see a clear increase in peak-ADC values over channels on link 0. To explain this phenomenon, we switch our focus to the mean-pedestal values of the time-scans, seen in the same Figure. Present across the entire CALIB range on link 0, a clear rising-pedestal pattern emerges, with the trend visually matching the peak-ADC increase. Curiously, the results of HGCROC board 22 closely resemble the pattern on the same link, but only on CALIB 256.

To investigate further, we performed additional scans on board 21 in reverse-order of channels - first scanning channel 35 and moving down to channel 0. If the emergent pattern matched the previous results, the likely culprit would be some type of inter-link amplification due to a software or hardware malfunction. However, the additional scans failed to replicate the results. The new data clearly indicated healthy behaviour on link 0. As the pre-CC time-scans of both board 21 and 22 were performed within a short period of time, it is likely that the issue stems from some failure in initial configuration.

Split sample-distributions Boards 20, 27 and 28 present a wholly unique type of irregularity: split sample-distributions. To illustrate, channel 0 on board 27 clearly shows nearly every phase of a pre-CC time-scan divided into two ADC-clusters (see Figure 18). Notably, these irregularities stay below the RMSD threshold despite their prominence. In these boards, the samples usually divide into two

neat clusters, so their average falls somewhere in the middle. This creates the false appearance of a healthy RMSD and mean pulse shape.

As this data was taken with both TDCs off, TOA/TOT triggers are an unlikely explanation to this behaviour. Instead, it is possible that these anomalies are caused by misconfiguration or malfunction of the trigger system, responsible for ensuring the reliability of single-sampling. Another possibility is misconfiguration of the shaper circuit, which, in part, ensures different pulses do not overlap during read-out. Due to the unique nature of this issue, it is difficult to identify it using link-wise RMSD. Instead, we suggest an automated identification algorithm described in Section 6.

Peak-ADC fluctuations Overall, we can see a general increase in link-wise RMSD values for both pre-CC and HR time-scans. Excluding results stemming from the ominous rising-pedestals of board 21 and 22, as well as the non-zero RMSD values after saturation, all of the tested boards follow this pattern. The pre-CC results of board 23 serve as a good example (see Figure 21). The peak-ADC values clearly fluctuate across the CALIB range. Checking the raw data, it is clear that these fluctuations are neither the result of pedestal misalignment nor any other aforementioned individual irregularity. Furthermore, the fluctuations get magnified with increasing CALIB values.

This anomaly is likely a result of innate manufacturing differences between the tested boards. Each channel of each board contains an individual PA circuit (see Section 4.2), which may amplify the signal with a slightly different gain. We would hence expect higher amounts of injected charge to correlate with higher peak-ADC deviation within each link. Small differences between capacitors in each channel may also induce variation in pulse-magnitudes. However, we would also expect capacitor differences to affect the link-wise RMSD values somewhat randomly, instead of only increasing them with pulse-magnitude, so they cannot fully account for the observed fluctuations.

HR		Link 0				Link 1			
Board/Peak-ADC-RMSD [ADC units]		CALIB 64	CALIB 128	CALIB 256	CALIB 512	CALIB 64	CALIB 128	CALIB 256	CALIB 512
Board 20		1.61	1.90	2.99	0	1.97	2.56	3.97	0
Board 21		2.16	2.70	4.40	0	1.66	1.90	3.08	0
Board 22		7.35	10.3	17.1	31.0	1.39	2.43	5.73	5.40
Board 23		2.71	2.58	5.21	3.37	2.39	3.68	3.42	6.04
Board 24		2.35	3.15	4.28	0	2.11	2.35	3.74	0
Board 25		38.8	50.8	36.9	13.2	2.74	3.46	4.94	0
Board 26		1.90	2.21	3.08	0	4.45	4.64	5.22	0
Board 27		2.21	2.84	4.42	0	1.98	2.26	3.09	0
Board 28		3.13	3.50	4.61	0	2.34	2.59	3.62	0
Board 29		1.81	2.42	4.12	0	2.56	2.93	4.12	0
Board 30		1.81	2.42	4.12	0	2.56	2.93	4.12	0

Table 2: The link-wise RMSD of mean-peak-ADC values for HR time-scans on HGCROC boards 20-30. The results are split by CALIB value and are calculated in ADC units. Furthermore, the RMSD values are rounded to 3 s.f., with values above 10 ADC units marked in red. As CALIB 512 is usually saturated, most of the peaks will consistently equal 1023 ADC units, resulting in null RMSD. The mean-ADC peaks are calculated from 50 samples on each channel, with the exception of 100 samples on HGCROC board 22.

Pre-CC		Link 0					Link 1				
	Board/Peak-ADC-RMSD [ADC units]	CALIB 256	CALIB 512	CALIB 1024	CALIB 2048	CALIB 4095	CALIB 256	CALIB 512	CALIB 1024	CALIB 2048	CALIB 4095
	Board 20	1.73	3.19	5.13	9.10	15.4	2.52	3.53	5.86	10.4	18.2
	Board 21	88.2	96.8	112	129	79.9	2.14	2.88	4.68	8.22	14.3
	Board 22	87.4	3.63	5.48	9.18	15.0	1.42	2.25	3.98	7.08	12.6
	Board 23	1.98	2.64	3.84	6.44	10.9	2.82	3.39	5.65	9.97	17.4
	Board 24	2.51	3.52	5.47	9.27	15.8	2.53	3.49	5.42	9.03	15.1
	Board 25	34.5	42.1	57.4	88.1	145	3.01	3.98	6.13	10.2	31.2
	Board 26	2.76	4.23	6.34	10.7	18.4	5.16	5.60	7.06	10.1	15.7
	Board 27	2.44	3.30	4.84	8.08	13.1	2.25	3.09	4.91	8.69	14.5
	Board 28	3.56	4.13	5.81	8.80	14.3	2.82	4.12	6.30	10.7	17.7
	Board 29	2.13	2.71	4.19	7.03	71.5	3.27	4.24	6.18	10.4	17.1
	Board 30	2.13	2.71	4.19	7.03	71.5	3.27	4.24	6.18	10.4	17.1

Table 3: The link-wise RMSD of mean-peak-ADC values for pre-CC time-scans on HGCROC boards 20-30. The results are split by CALIB value and are calculated in ADC units. Furthermore, the RMSD values are rounded to 3 s.f., with values above 10 ADC units marked in red. The mean-ADC peaks are calculated from 50 samples on each channel, with the exception of 100 samples on HGCROC board 22.

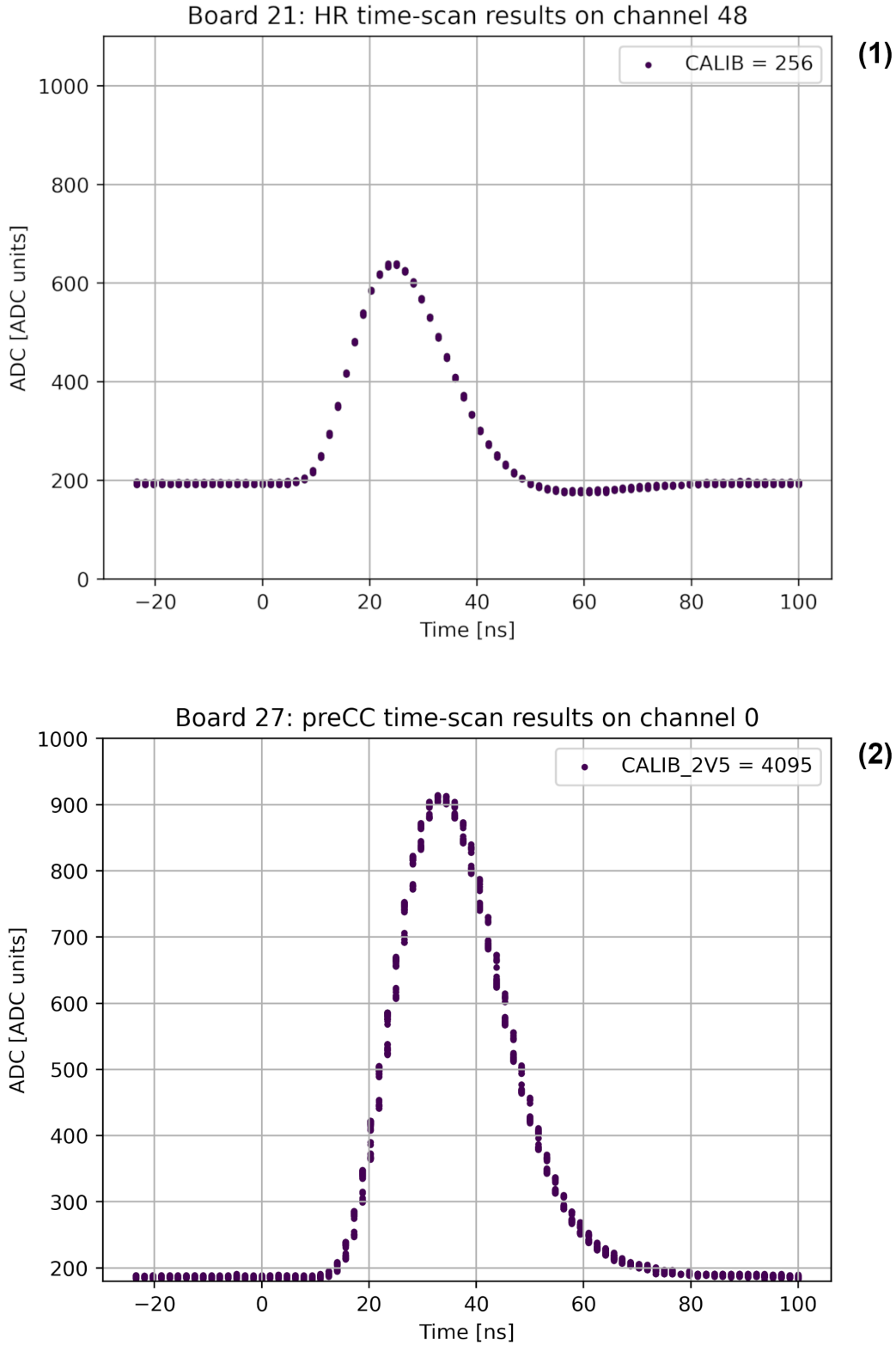


Figure 18: Plot of a healthy HR time-scan response on HGCROC board 21 (1) and unhealthy pre-CC response on board 27 (2). Both figures include 50 samples per phase, taken at CALIB 256 on channel 48 (1) and CALIB 4095 on channel 0 (2). (1): We note the tight sample-distributions on each phase, rarely wider than a couple ADC units. (2): The sample-distributions at each phase after the pedestal-level are noticeably split into clusters.

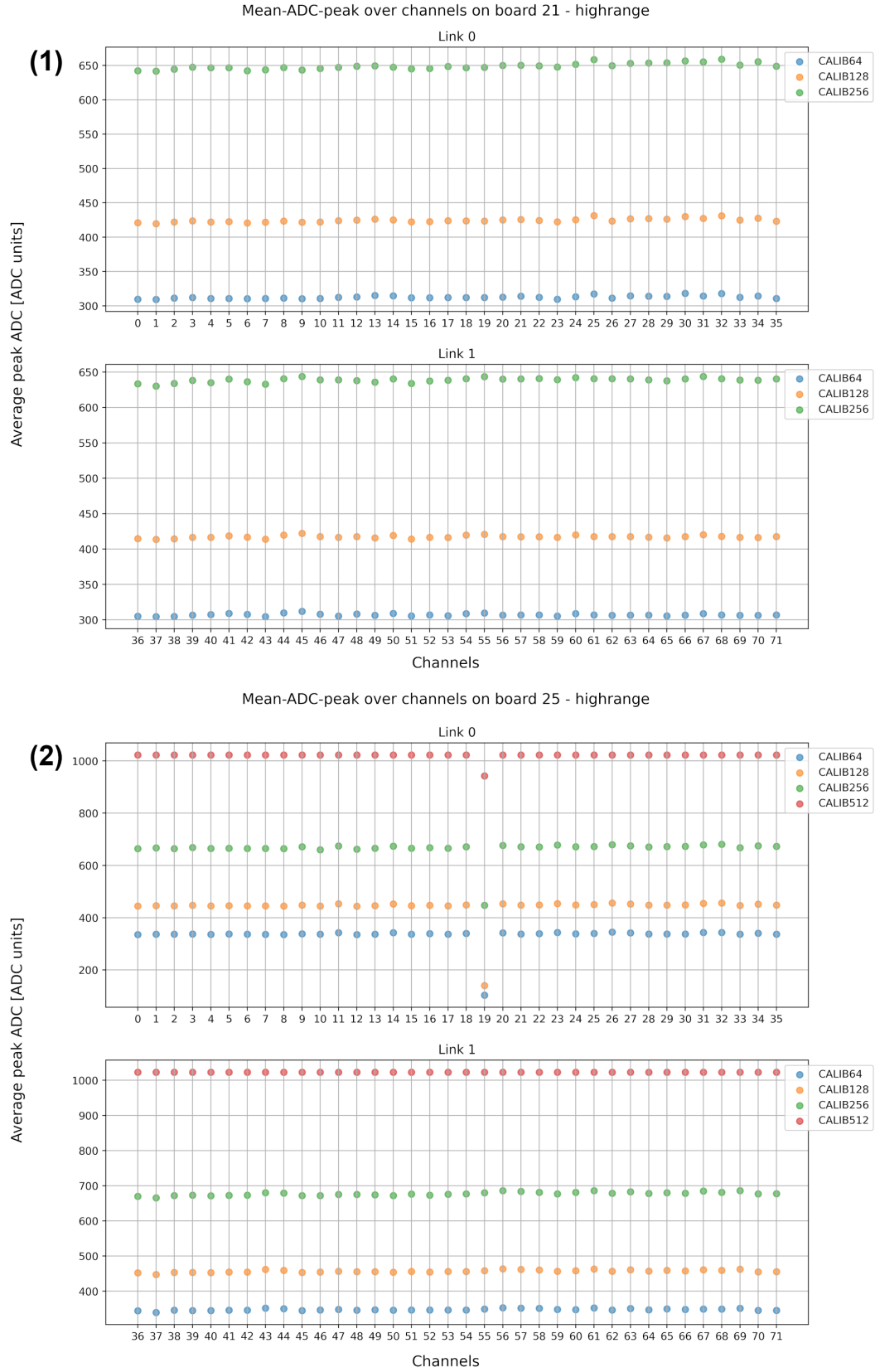


Figure 19: Plot of HR mean-peak-ADC values on HGCROC board 21 **(1)** and board 25 **(2)**, across multiple CALIB values. The means are taken over 50 samples on each channel. Each figure is split into plots of link 0 and link 1. The corresponding plots of channel-wise RMSD results can be found in Appendix E.2. **(2)**: We note the substantial dip on channel 19 for all CALIB values.

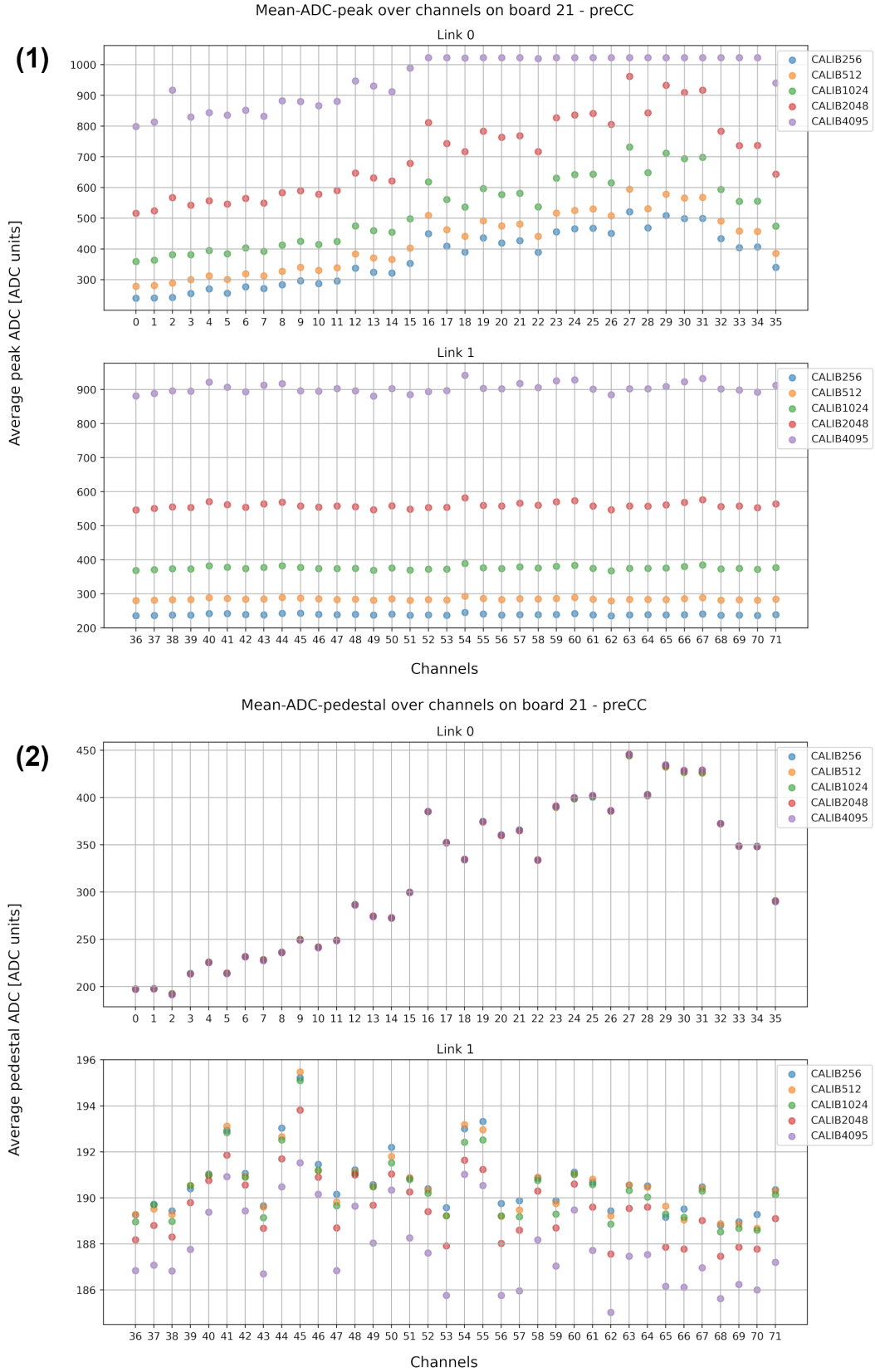


Figure 20: Plot of pre-CC mean-peak-ADC (1) and mean-pedestal-ADC (2) values on HGCROC board 21 across multiple CALIB values. The means are taken over 50 samples on each channel. Each figure is split into plots of link 0 and link 1. We note the unanimity of pedestal values in (2), translating into a very similar trend in (1), across all CALIB values.

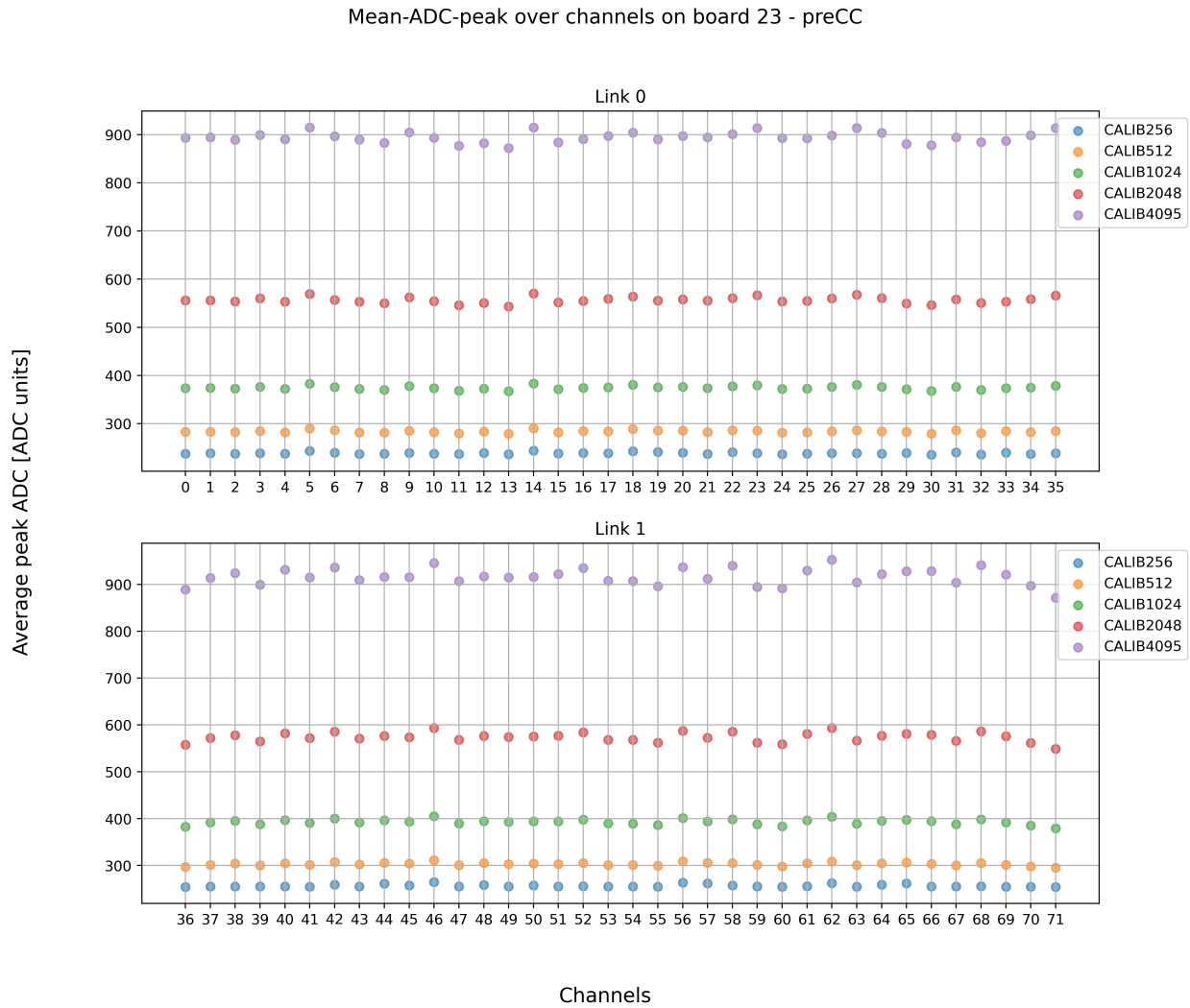


Figure 21: Plot of pre-CC mean-peak-ADC values on HGCROC board 23 across multiple CALIB values. The means are taken over 50 samples on each channel. The top figure shows link 0, while the bottom figure link 1. We note the fluctuations seen on both links, particularly at the high CALIB values.

5.4.3 ADC/TOT linearity

We evaluate the ADC/TOT linearity by tracking the R^2 -value of linear fits applied to peak-ADC and TOT values taken with an HR charge-injection over a large CALIB range. The fits are applied to peak-ADC values before saturation and the TOT values in the linear TOT regime. The R^2 -value ranges from 0 to 1, with an R^2 -value of 1 suggesting the data perfectly maps onto the linear fit. As can be seen in Figures 23 and 24, all of the HGCROC boards, with the exception of board 21, show near-perfect linear behaviour on both peak-ADC and TOT fronts.

TDC malfunction While HGCROC-board 21 presents linear results for peak-ADC values on both links, it fails to replicate them for TOT values on link 1. As marked in the description of Figure 24, the null- R^2 values do not indicate non-linear TOT behaviour. Instead, the data indicates both TDCs are completely inactive throughout the scans.

TOT gap Although the TOT linear regime adequately lives up to its name with perfect R^2 -results, it is important to note the location of said regime. As the TOT values rise with increasing injected charge, we expect a short period of non-linear growth before entering a linear TOT region. We can use this linear domain for increasing the measurable ADC range. Instead, as can be seen in Figure 22, the linear TOT region is entirely disconnected from its initial curve, resulting in a large gap. The **TOT gap** is present in *all* CALIB-scans, and varies in both CALIB and TOT width. Though somewhat inconsistent, the linear region was observed to always appear after at least 550 TOT units, allowing for the region to be selected for the linear fit. As this anomaly could lead to substantial issues during the experiment, we suggest further study into the HGCROC board configuration and TDC-response.

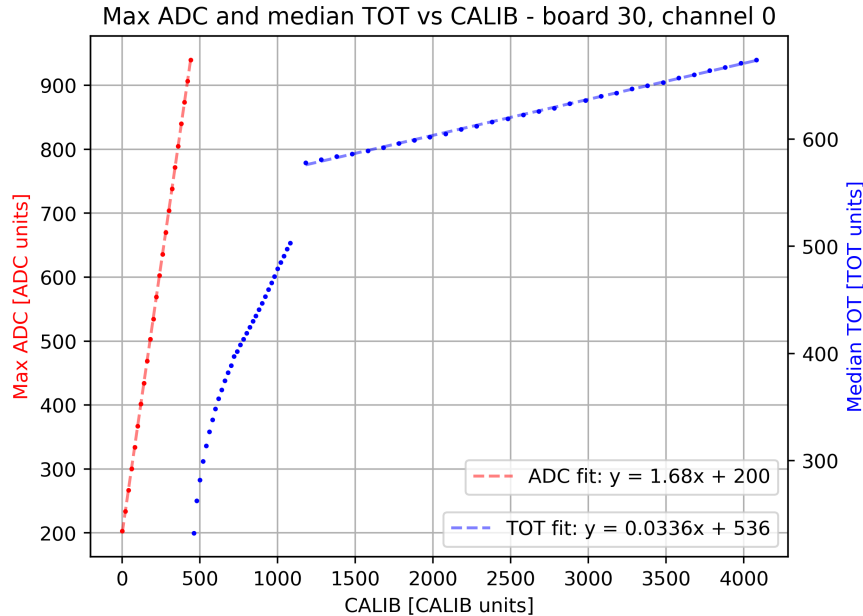


Figure 22: Plot of max (peak) ADC and median TOT values across CALIB values 0-4080. The max ADC and median TOT were taken as the median of 10 samples. The max ADC shows linear behaviour until saturation, while the median TOT values present a slow ramp-up leading to a linear trend, separated by the TOT gap. The ADC and TOT fits can be seen in the lower left corner.

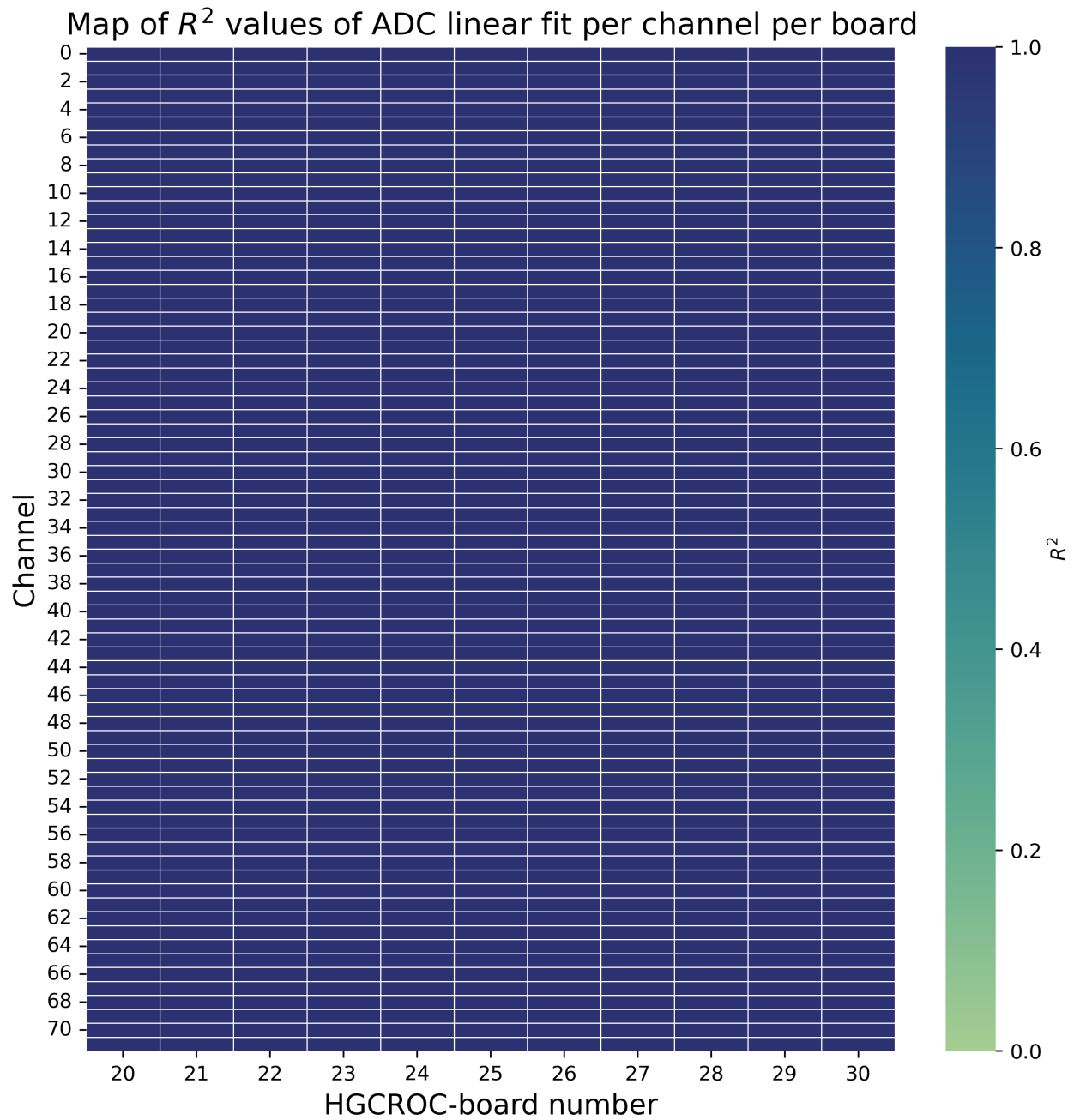


Figure 23: R^2 values of ADC linear fits across channels and HGCROC boards. An R^2 value of 1 indicates the data perfectly aligns with the fit. All of the tested boards show R^2 values of around 0.99.

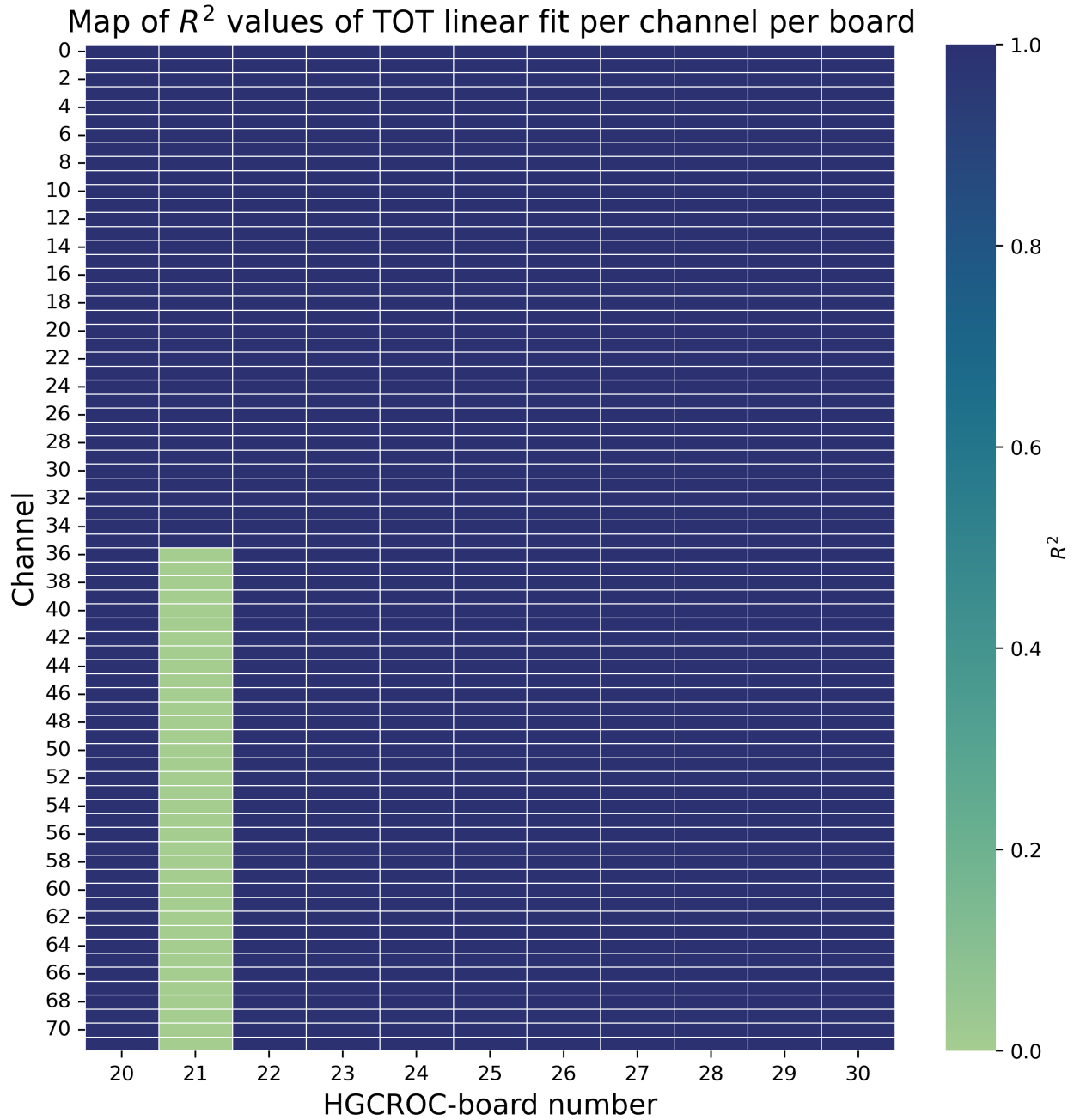


Figure 24: R^2 values of TOT linear fits across channels and HGCROC boards. An R^2 value of 1 indicates the data perfectly aligns with the fit. All of the tested boards, except board 21, show R^2 values of around 0.99. As the TDCs of board 21 did not activate during measurement on link 1, we present the inactive link as $R^2 = 0$.

5.4.4 TOA response

As described in Section 5.3.1, we evaluate the TOA response qualitatively, focusing on median-TOA time-walk plots, with the median taken from 10 samples across TOA-active phases on each channel. Fortunately, most of the irregular TOA responses are fairly obvious. As an example, HGCROC-board 23 (see Figure 25) shows an expected TOA evolution on link 0, and a somewhat irregular response on link 1, with small median-TOA spikes around between CALIB 0 and 1000. In contrast, the TOA values of board 20, seen in Figure 26, are very chaotic, likely due to sample ADC-irregularities or possibly malfunctioning TOA-TDC. Board 20’s behaviour is fairly unique. The rest of the time-walk irregularities fit one of several categories⁸:

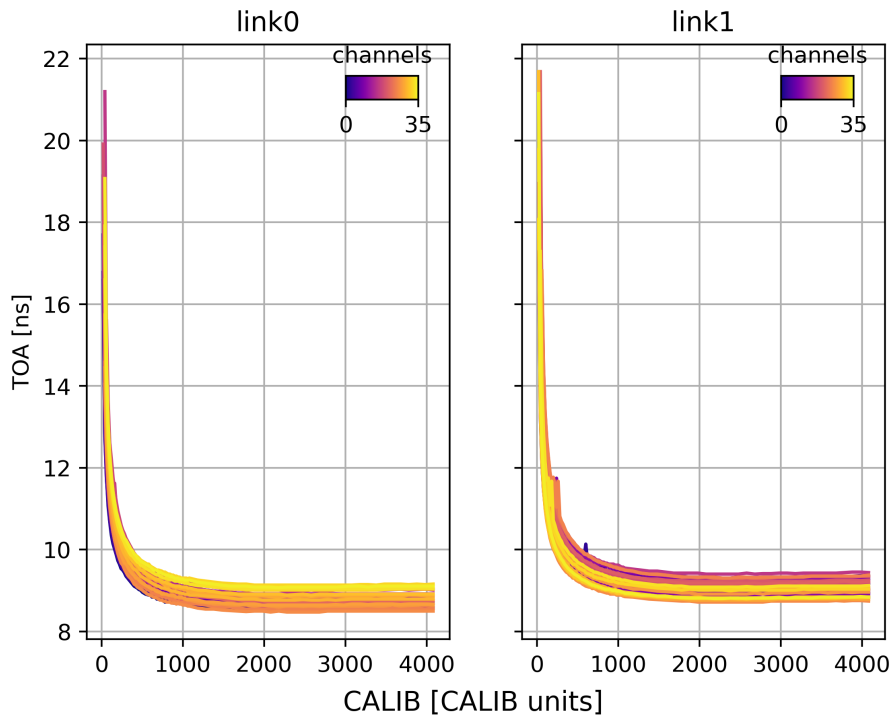


Figure 25: TOA time-walk taken via CALIB-scans on board 23. The data spans CALIB values 0-4080 on both links. As a pulse increases in magnitude, the TOA-TDC triggers at earlier phases. We note the small spikes on link 1.

TOA-jumps As can be seen on link 0 of board 21 (see Figure 27), a common unhealthy TOA response is sudden jumps in median-TOA values across the time-walk. This issue presents in virtually all tested boards, even overlapping with other irregularities. The jumps are likely caused by a fault in the TOA-threshold-setting `pflib`-procedure, and require further study.

TOA-TDC malfunction Providing an explanation for the lack of TOT values in Section 5.4.3, HGCROC board 21 shows no TOA response on link 1 (see Figure 27). Though unique to board 21, we delineate this irregularity due to its lack of consistency. The TOA response of board 21 was probed on multiple different occasions, with the null link-1 TOA manifesting seemingly on a whim.

⁸Notably, the described anomalies tend to only appear in the lower-CALIB range, leaving high-CALIB TOT-values unaffected, yielding positive TOT-linearity results.

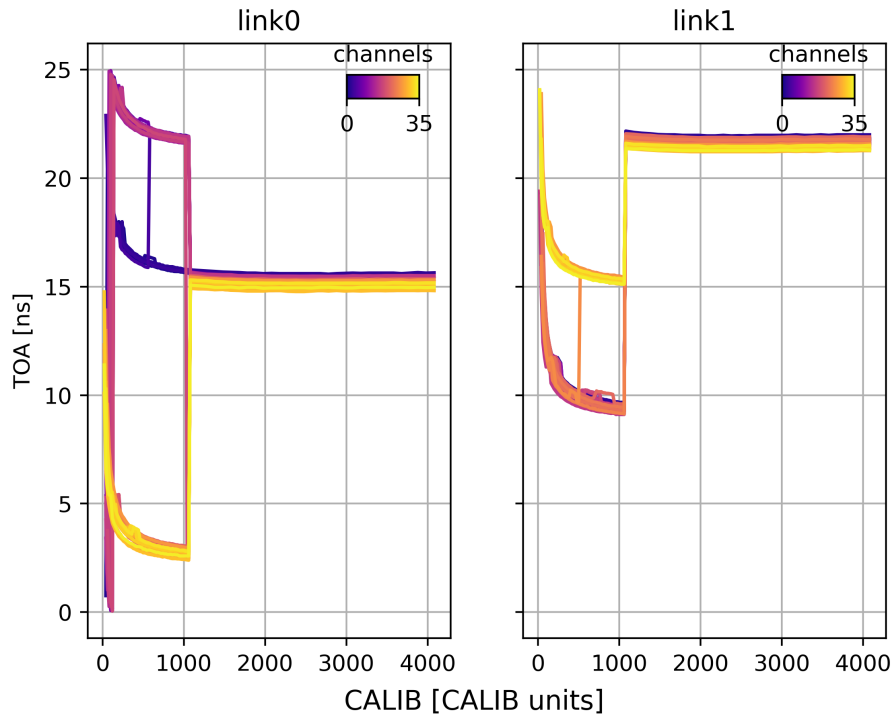


Figure 26: TOA time-walk taken via CALIB-scans on board 20. The data spans CALIB values 0-4080 on both links. We note the clearly chaotic behaviour.

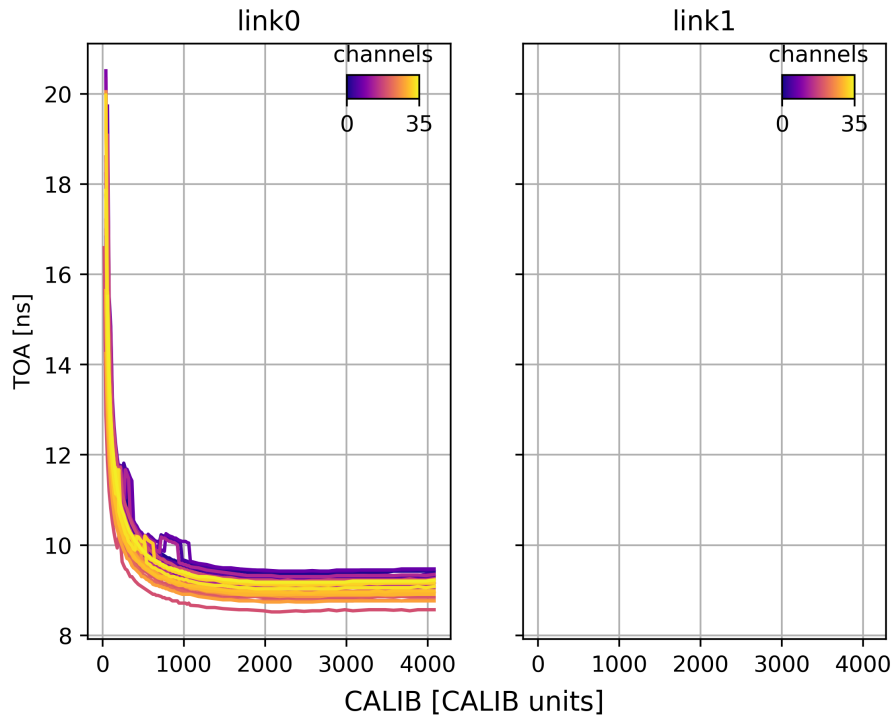


Figure 27: TOA time-walk taken via CALIB-scans on board 21. The data spans CALIB values 0-4080 on both links. We note the TOA-jumps on link 0 and a lack of TOA response on link 1.

Low-CALIB spikes Following the TOA response shown on Figure 28, we see an abrupt rise in TOA in the lower-CALIB region. This behaviour shows up on HGCROC-boards 27 and 29, across both links and link 1, respectively. Though alarming at first glance, this response can be easily explained by imagining the evolution of a small pulse. As a small amount of charge is injected on a channel, it is possible for the pulse to miss the TOA threshold in the first bx (25 ns), as illustrated in Figure 29. However, the pulse is still likely to reach the threshold at the beginning of the *next* bx. As the TOA-TDC measures the amount of time passed between the beginning of a bx and a pulse reaching the threshold, the TOA value for such small pulses would be measured with respect to the second bx. This would result in an incredibly short, near-zero TOA, as we can see in Figure 28. As the pulse-magnitude increases, the TOA threshold is reached at the end of the first bx, thus resulting in a value close to 25 ns. In short, it is very likely that, for boards 27 and 29, the TOA threshold was set slightly too high for small charge-injections, resulting a sudden TOA spike.

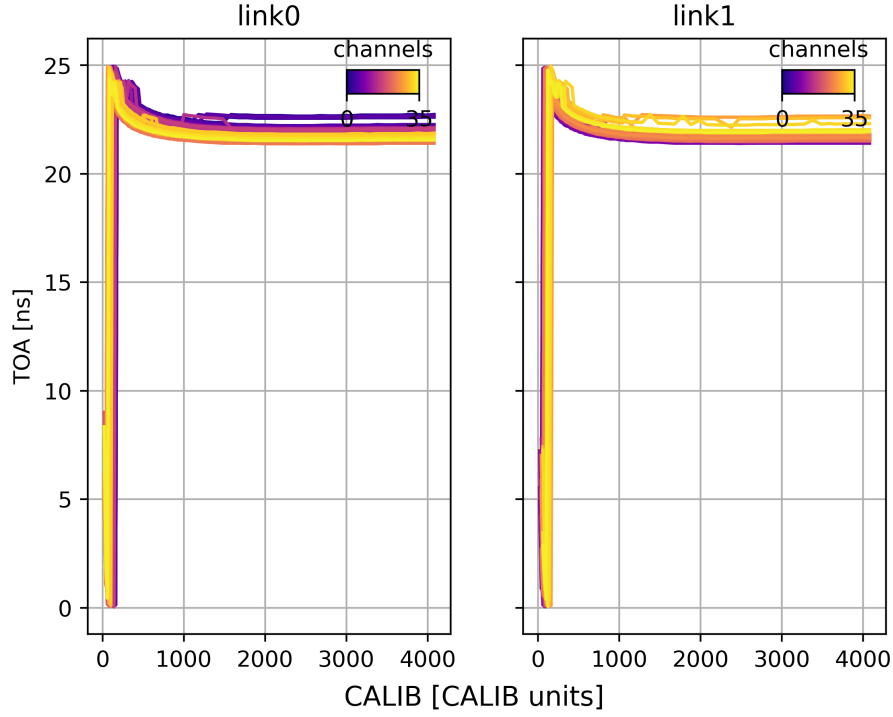


Figure 28: TOA time-walk taken via CALIB-scans on board 27. The data spans CALIB values 0-4080 on both links. We note the sudden spikes in TOA value in the lower-CALIB region on both links.

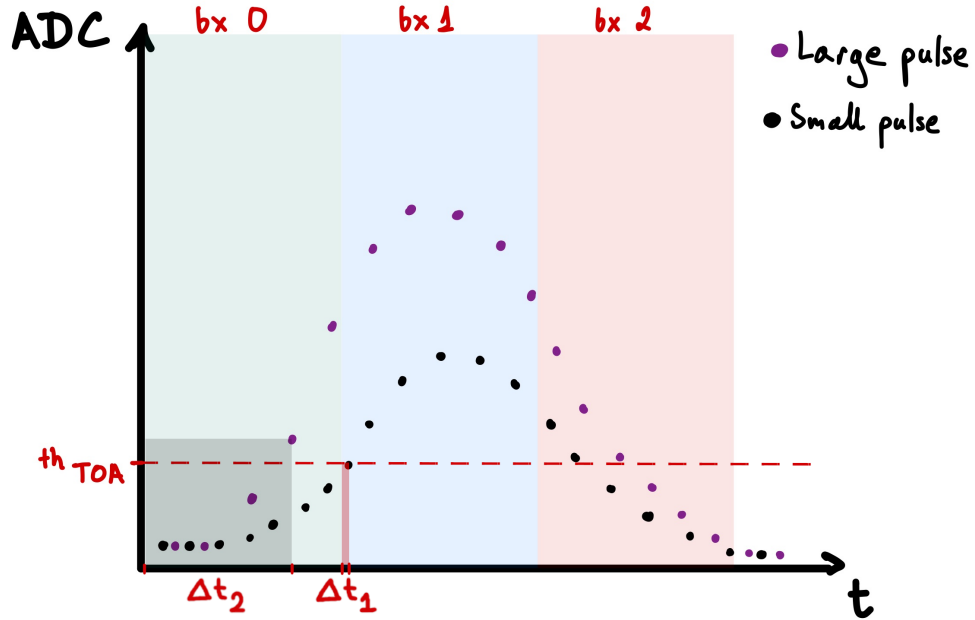


Figure 29: Sketch of two pulses triggering the TOA-TDC at different bxs. The pulses and bx widths are not to scale. The large pulse (purple) passes the TOA-threshold (th_{TOA}) at bx 0, resulting in a large Δt_2 . In contrast, the small pulse (black) only passes th_{TOA} at the beginning of bx 1, leading to a very small Δt_1 .

Board/Irregularity	Board 20	Board 21	Board 22	Board 23	Board 24	Board 25	Board 26	Board 27	Board 28	Board 29	Board 30
Pedestal alignment						x	x		x	x	
Peak-ADC alignment			x	x	x					x	
		x	x								
	x							x	x		
	x	x	x	x	x	x	x	x	x	x	x
ADC/TOT linearity	x	x	x	x	x	x	x	x	x	x	x
TOA response		x			x	x	x	x	x	x	x
	?	x						x			

Table 4: Summary table of the various irregularities in the data analysis results of the tested HGCROC boards found in Section 5.4. The irregularities are marked with an **x**. TOA/TOT triggers and peak-ADC fluctuations are coloured grey, as they do not reflect the health of the HGCROC boards. We note the **?** symbol in the TOA-TDC malfunction of board 20, indicating inconclusive diagnosis. Due to multiple overlapping issues and the unique shape of the TOA time-walk, the TOA response of board 20 could be attributed to TDC malfunction, as well as various other sources.

6 Sample-clustering Identification Algorithm

6.1 Introduction

As described in Section 5.4, split clusters of time-scan samples constitute a specific class of charge-injection irregularities (see Figure 18). We found that the peak-ADC alignment evaluation metric tended to miss this type of anomaly. As such, we devised an automated sample-clustering identification algorithm as a supplemental tool. In this section, we list the various features of the algorithm, alongside potential improvements.

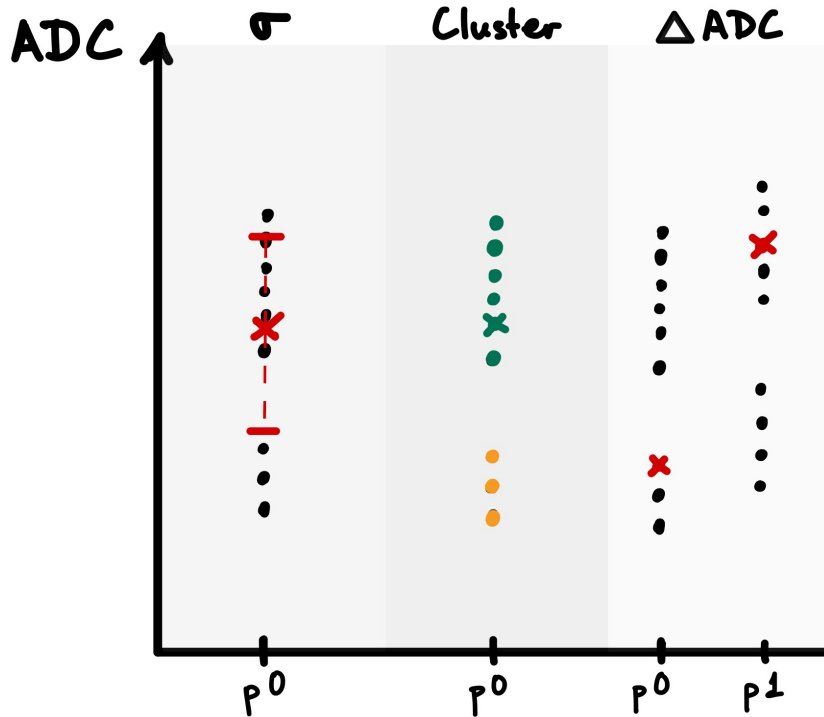


Figure 30: Sketch of different filtering methods. Each method is illustrated using ADC sample-distributions on one or two phases (p). On the left, σ -based filtering locates the mean ADC value (marked with ‘x’), and labels all samples outside at least 3σ from the mean (marked with dashes) as outliers. In the middle, **cluster**-based filtering locates the mode (marked with ‘x’), and divides the sample-distribution into clusters, highlighted with different colours. On the right, ΔADC -based filtering selects samples (marked with ‘x’) by their measurement order, e.g. third sample of each phase, and takes their ADC-value difference.

6.2 Algorithm versions

The identification algorithm was chosen from three different versions, each using a unique filtering method. After a visual assessment, we found that the pre-CC time-scan data taken on board 23 at CALIB value 4095 was largely free of split sample-distributions. These data were hence used for finding the false-positive rate of each algorithm. We define false-positives as samples labelled anomalous, while not clearly disconnected from a tight, ‘true’ sample-distribution, or whose variation is indistinguishable from natural ADC fluctuations. The versions are based on the following methods (illustrated in Figure 30):

Δ ADC-based filtering: This method monitors the differences in ADC values (Δ ADC) of subsequent phases of a reconstructed ‘sample-pulse’. The samples first categorized by their measurement order within each phase. They are then coalesced into a single sample-pulse. For example, each third ADC-sample of every phase is taken out to a separate array, making up the third sample-pulse. Then, each phase is assigned a Δ ADC value, taken by subtracting the ADC value of the previous phase from its ADC value. The first phase is assigned a Δ ADC value of 0.

σ -based filtering: Instead of treating samples as part of the same pulse, this method focuses on the distribution of ADC values at each phase, and identifies outliers based on the standard deviation (σ). Considering each phase one-by-one, the algorithm finds the sample-distribution mean and σ . Samples are marked as outliers when their corresponding ADC value is at least 3σ away from the mean.

Cluster-based filtering: The final version of the identification algorithm is designed to divide the sample-distributions at each phase into clusters. First, the algorithm finds the mode (most frequent ADC value) of the sample-distribution. The script then splits into two loops, each moving up and down by one ADC unit from the mode. Each loop marks samples with equal ADC values. If no sample is found within a range of two to five ADC units from the last identified sample, the algorithm marks the last sample as the lower/upper-end of the ‘true’ cluster (containing the mode). The samples after that gap are then marked as ‘potential outliers’. Similarly, when no sample is found for more than five ADC units, the samples after the gap are marked as ‘outliers’. Furthermore, any new clusters found after the (potential) outlier-cluster are marked as ‘outliers’ themselves. Gaps of one ADC unit are ignored to account for natural fluctuations.

Table 5 presents an estimate of the false-positive rate of each method. This estimate is derived as the average proportion of samples containing any identified anomalies per channel, within each link. First, we address the seemingly perfect results of the Δ ADC-based algorithm. This method requires a manually chosen Δ ADC-threshold, hand-tuned for each CALIB. This process is obviously inefficient and time-consuming. We found that the method had a null or 100% false-positive rate, making it the least useful identification method. The cluster and σ -based algorithms provide more interesting results. Though σ -based filtering often succeeds in avoiding false-positive identification above pedestal-level, pedestal fluctuations overwhelm it. Furthermore, the method often marks samples on the edges of their distribution as irregularities. This is to be expected, as σ adjusts to any given sample-distribution, often ensuring irregularities are identified even in very tight clusters. In contrast, the cluster-based algorithm shows a remarkably low false-positive rate. This is likely due to the distinction between outliers and potential outliers, which notably hold much higher mean-fractions on each link. As such, we have chosen cluster-based filtering as the primary identification method used by the algorithm.

6.3 Analysis tools

The sample-clustering identification algorithm comprises a variety of data-analysis tools, which help diagnose the roots of notable charge-injection ADC value irregularities. Though the algorithm uses cluster-based filtering by default, the other two methods are also available. The most important features are listed as follows:

Mean fraction [ADC units]/Filtering-method	Cluster	σ	ΔADC
Outlier-samples (link 0)	0.071	0.398	0
Outlier-samples (link 1)	0.030	0.299	0
Potential outlier-samples (link 0)	0.608	-	-
Potential outlier-samples (link 1)	0.204	-	-

Table 5: Average fraction of samples with identified sample-irregularities in a pre-CC CALIB 4095 time-scan on board 23. The averages are calculated per link, with each channel containing 50 samples. σ and ΔADC -based methods do not distinguish potential outlier-samples. The fractions are listed up to 3 s.f. in ADC units.

Phase analysis This tool is designed for phase-categorization of identified anomalies, currently supporting one channel and CALIB value at a time. As the time-scan data is saved with time in ns, each time-point is converted into its corresponding phase within a bx. Using this function, the user is able to determine and plot the outlier-sample phase distribution (see Appendix H), which can prove useful during malfunction diagnostics.

Board-wide analysis Though the algorithm is most useful for detailed, channel-by-channel evaluation, we include the ability to perform board-wide and CALIB-wide frequency analysis. The ‘bulk-evaluation’ function iterates through all the provided time-scan data files, storing the total number of identified irregularities and the fraction of samples that contain them. This data can then be exported to plotted.

Plotting functionality The script contains a range of plot types, separate from phase-analysis figures, illuminating the characteristic features of identified outlier-samples. The data can be viewed as sample-pulses, with marked irregularities. Furthermore, the samples can be plotted all at once, allowing for visual assessment of split sample-clusters. Lastly, the results of the board-wise analysis can be plotted as histograms. All of the aforementioned functions are available for all filtering methods and outlier-sample types (see Appendix H for examples).

6.4 Future improvements

Although cluster-based filtering performed the best out of the three options, the result is fairly biased. As noted in Section 5.4, ADC value fluctuations tend to increase with higher CALIB values. Given that this method uses rigid thresholds for identification, we may expect it to falter when it comes to drastically different CALIB values. As such, future improvements to the algorithm should focus on making the thresholds malleable. This could be done by first characterising the ADC value fluctuations of tested HGCROC boards. Using this information, an automated script could be used for tuning the thresholds to an acceptable false-positive rate for charge-injections at any given CALIB value.

7 Conclusion & Outlook

This study presents a comprehensive evaluation of ten H2GCROCv3b boards, using a new automated testing procedure. The internal charge-injection tests were analysed with respect to four metrics, which reflect the reliability of the chip-board read-out.

Examining pedestal alignment, we found that the read-out of the tested boards was largely reliable. However, the results still suggest improvements to the `pftool` pedestal-levelling and lowering functions are necessary before the final experiment.

The evaluation of peak-ADC alignment yielded many irregularities in the charge-injection response of the HGCROC boards. While some, like TOA/TOT triggers or increasing pedestals, were due to errors in the initial setup, others could not be similarly dismissed. We found that the aforementioned pedestal-alignment heavily affected the evaluation results. Furthermore, we noticed concerning behaviour at higher CALIB-ranges, namely visible peak-ADC fluctuations. Though likely explained by manufacturing differences between different electronic components of the ASIC and/or chip board, we believe this issue requires further investigation. Lastly, divided time-scan sample-distributions were found in three boards, and could not be explained using the available data. To pave the way for future study of this issue, we have constructed a sample-clustering identification algorithm, comprising various features tailored for split sample-distribution analysis.

Although mostly positive, the results of the ADC/TOT-linearity evaluation also marked some notable read-out anomalies. We found that one board contained potentially malfunctioning TDCs, affecting only link 1. What we found most concerning was this feature's transient nature - showing itself one day, only to disappear on another. The TOT gap proved a more consistent anomaly, presenting as a visible disconnect between the non-linear and linear parts of the TOT-curve. As the TOT is a crucial part of the HGCROC read-out, this issue requires deeper study.

Finally, the TOA-related setup procedures clearly need further development. Though only qualitatively, we found multiple irregularities in the TOA time-walk curves on each board. Some, like the low-CALIB spikes, were clearly a result of an improper TOA-threshold, which is entirely dependent on the `pftool` setup functions. Others, such as the TOA-jumps, could not be easily explained, and were found on almost all of the tested boards. Overall, the TOA response seems like the least understood part of the HGCROC read-out, providing ample room for investigation.

Though only covering a fraction of the H2GCROCv3b functionality, we hope that this study can help in further development of diagnostic tools as part of the `pflib` control software, as well as improve the reliability of the H2GCROCv3b ASICs in preparation for LDMX.

References

- [1] G. Bertone and D. Hooper, “History of dark matter,” *Reviews of Modern Physics*, vol. 90, no. 4, Oct. 2018, ISSN: 1539-0756. DOI: 10.1103/revmodphys.90.045002. [Online]. Available: <http://dx.doi.org/10.1103/RevModPhys.90.045002>.
- [2] M. Cirelli, A. Strumia, and J. Zupan, “Dark matter,” *SciPost Phys. Rev.*, p. 1, 2026. DOI: 10.21468/SciPostPhysRev.1. [Online]. Available: <https://scipost.org/10.21468/SciPostPhysRev.1>.
- [3] Gajdán, Greta, *LDMX HCal prototype data analysis*, eng, Student Paper, 2025.
- [4] C.-Y. Xing and S.-h. Zhu, “Dark matter freeze-out via catalyzed annihilation,” *Phys. Rev. Lett.*, vol. 127, p. 061101, 6 Aug. 2021. DOI: 10.1103/PhysRevLett.127.061101. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.127.061101>.
- [5] T. Akesson et al., *Ldmx – the light dark matter experiment*, 2025. arXiv: 2508.11833 [hep-ex]. [Online]. Available: <https://arxiv.org/abs/2508.11833>.
- [6] S. Navas et al., “Review of particle physics,” *Phys. Rev. D*, vol. 110, p. 030001, 3 Aug. 2024. DOI: 10.1103/PhysRevD.110.030001. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.110.030001>.
- [7] M. Fabbrihesi, E. Gabrielli, and G. Lanfranchi, *The Physics of the Dark Photon: A Primer*. Springer International Publishing, 2021, ISBN: 9783030625191. DOI: 10.1007/978-3-030-62519-1. [Online]. Available: <http://dx.doi.org/10.1007/978-3-030-62519-1>.
- [8] T. Markiewicz, T. Raubenheimer, N. Toro, and members of the LESA construction team, *The slac linac to esa (lesa) beamline for dark sector searches and test beams*, 2022. arXiv: 2205.13215 [physics.acc-ph]. [Online]. Available: <https://arxiv.org/abs/2205.13215>.
- [9] T. Åkesson et al., *Photon-rejection power of the light dark matter experiment in an 8 gev beam*, 2023. arXiv: 2308.15173 [hep-ex]. [Online]. Available: <https://arxiv.org/abs/2308.15173>.
- [10] T. Nelson, *Light dark matter experiment: A discovery experiment for sub-gev dark matter*, Accessed: 15-05-2026. [Online]. Available: https://indico.bnl.gov/event/18372/contributions/75188/attachments/46958/79635/LDMX_P5_Nelson.pdf.
- [11] J. González-Martínez, “H2gcroc: Design and performance of a dedicated very front-end asic for sipm readout of the cms high granularity calorimeter,” *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, vol. 1047, p. 167863, 2023, ISSN: 0168-9002. DOI: <https://doi.org/10.1016/j.nima.2022.167863>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S016890022201155X>.

Appendices

A Root-mean-square deviation

For the purposes of this study, we calculate RMSD as we would standard deviation⁹, i.e.

$$\text{RMSD} = \sqrt{\frac{1}{n} \sum_i^n (x_n - \bar{x})^2},$$

where n corresponds to the number of samples, x the sample value and $\bar{x}$ the mean-sample value. In the case of pedestals, for example, we calculate two types of RMSD. Taking the pedestal-samples in one channel, the RMSD denotes the maximum spread from the sample-mean ADC, indicating the health of the channel. On the other hand, the RMSD of the mean-pedestals within one Link denotes the spread from the Link-mean ADC, as if they are samples of a pedestal on one large channel.

B R^2 coefficient

During ADC/TOT-linearity data analysis, we perform linear regression via the `Python` `scipy.stats.linregress` function. One of the function's output is the r -value, or the Pearson correlation coefficient. We use the square of this value to obtain the R^2 coefficient of determination, which measures the correlation between the data and the fit using a scale from 0 to 1. The R^2 -value can also be derived as follows:

$$R^2 = 1 - \frac{\sum_i (y_i - f_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where y , f and $\bar{y}$ refer to the measured ADC/TOT value, the fit-ADC/TOT value, and the average ADC/TOT value, respectively.

C Testing procedure

The testing procedure is split into configuration and data acquisition, described in Section 5.3.2. In this appendix, we give a detailed list of the specific `pftool` functions used in the procedure, as well as the setup-functions that preceded it. The locations and names of the functions are listed as seen at the time of writing.

Setup

ROC/HARDRESET: Performs a hard-reset of the HGCROC board, reverting it back to the default configuration.

ROC/SOFTRESET: Additional reset to ensure the board is fully 'wiped'.

EXPERT/ELINKS/AUTO: Automatic setup of the electronic links of the chip board.

⁹Although standard deviation is a more common error measure, it can misleadingly imply a standard distribution of the data in question. As none of the considered HGCROC responses follow a standard distribution, we opt to use 'RMSD', inline with common signal-processing nomenclature.

ROC/RUNMODE: Sets up the chip for data acquisition.

DAQ/SETUP/STANDARD: Sets up the DAQ configuration to ‘standard’, as defined in `pflib`.

Configuration and data acquisition

The configuration and data acquisition were performed in order.

TASKS/LEVEL PEDESTALS: Responsible for pedestal alignment, this function configures each of the 72 HGCROC channels to produce a similar background-noise ADC response.

DAQ/PEDESTAL: This function records the read-out of the ADC while no charge is being injected, effectively taking measurements of background noise. For each ASIC, this scan was taken with 100 samples per channel. These pedestals are taken for a before- and after-lowering comparison (making sure the next script works properly).

TASKS/INV_VREF_SCAN_LUND: Sets specific reference voltages within the shaper-circuit on each channel to lower their pedestal values to a target threshold (around 200 in the Lund testing setup). The lower the pedestals, the wider the measurable ADC range. However, pedestals that are too low are more susceptible to noise-induced fluctuations.

TASKS/LEVEL PEDESTALS: The pedestals are levelled again after lowering.

DAQ/PEDESTAL: The pedestal data is taken again after levelling. This data is the final data used in pedestal-alignment evaluation.

TOA_VREF_SCAN: Sets the reference voltage for the titular TOA-threshold for each channel of the chip.

TOT_SCAN: Similarly to the function above, this scan configures the TOT reference voltage, which marks the pulse falling below a chosen threshold.

CHANNEL_WISE_CALIB_SCAN: This function measures ADC, TOA-TDC and TOT-TDC response after an HR charge-injection. The scan covers two bxs over a pre-defined 1100-4180 CALIB range, and was taken with 10 samples per channel on each chip.

PARAMETER_TIME_SCAN: This scan is used to collect ADC data over a chosen number of bxs, with the channels scanned, injection mode and CALIB value comprising the titular parameters. In this study, all of each chips’ channels were scanned in HR and pre-CC mode over four bxs. The data was taken for seven notable CALIB values, each measured with 50 samples per channel.

D `pflib` analysis scripts

To perform the data analysis described in Section 5.4, we developed multiple `pflib` Python scripts, available at the `pflib` GitHub repository. The final versions of the scripts can be found via the following links:

Pedestal averages: Used for all pedestal-alignment analysis and plotting. Available at <https://github.com/LDMX-Software/pflib/pull/408/changes> (averagesplot.py).

Time-scan-evaluation: Used for all peak-ADC-alignment analysis and plotting, with the exception of single-channel plots. Available at <https://github.com/LDMX-Software/pflib/pull/411/changes> (`time-scan-evaluation.py`).

Outlier Identification Algorithm: Described in detail in Section 6. Available at <https://github.com/LDMX-Software/pflib/pull/417/changes> (`outlier_scan.py`).

Scripts for TOA/TOT-related data analysis and plotting were primarily developed by Andreas Pettersson and Verena Hehl, and are available at the same repository.

E Supplementary figures

Below we include some supplementary figures for Section 5.4.

E.1 Pedestal alignment

Board 22: Mean pedestals and their RMSD on link 1

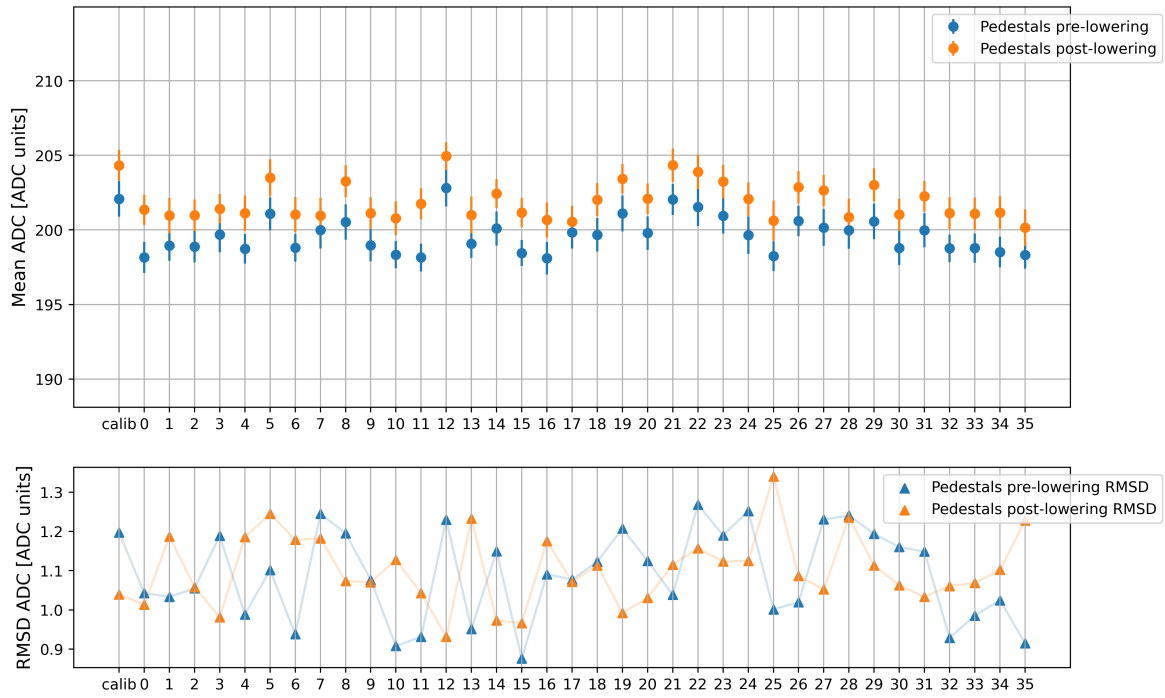


Figure 31: Plot of mean pedestals across link 1 on HGCROC board 22, before and after lowering. The means are taken over 100 samples on each channel. The top figure shows the mean pedestals with channel-wise RMSD error bars, while the bottom figure shows the channel-wise RMSD values.

E.2 Peak-ADC alignment

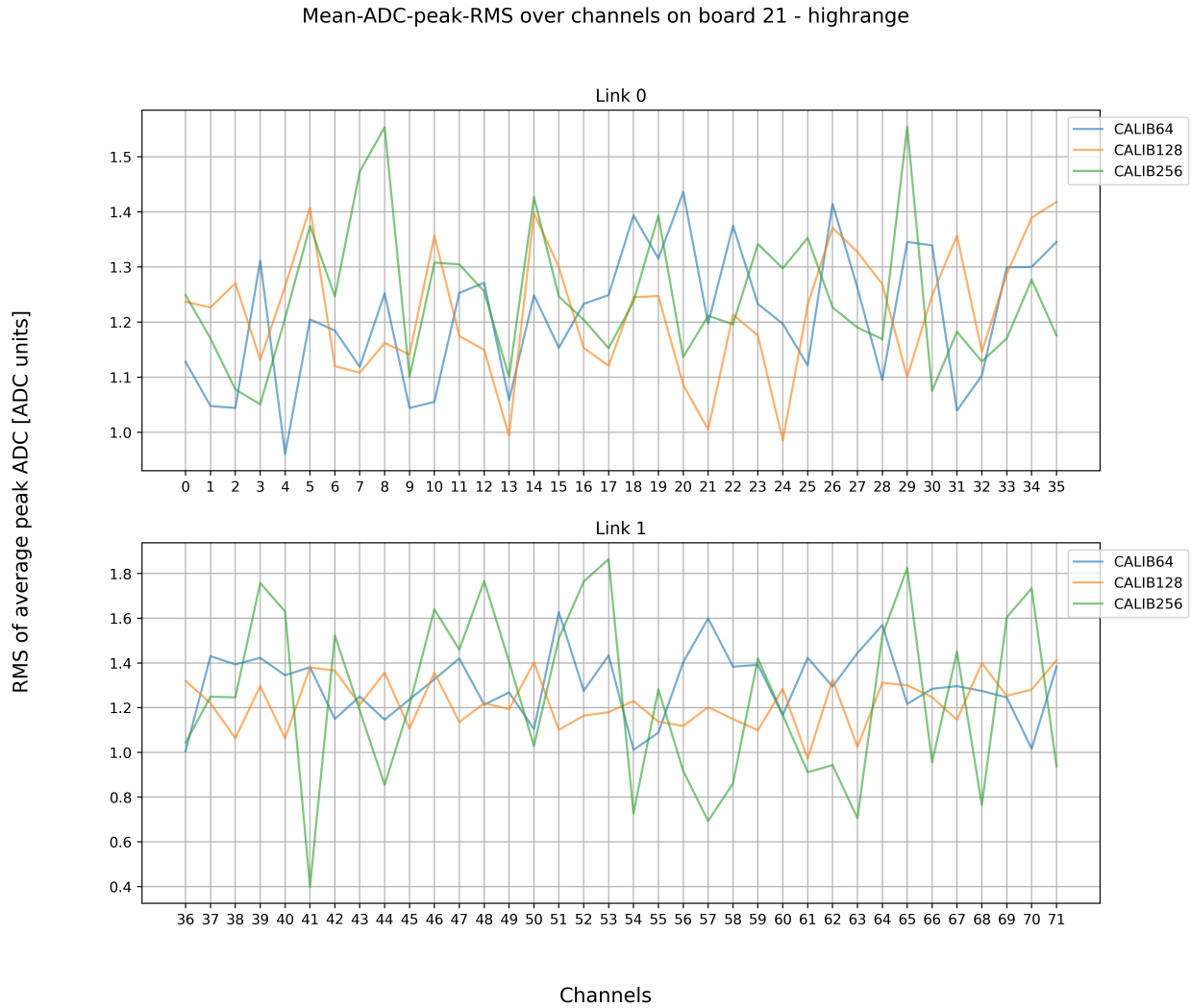


Figure 32: Plot of HR channel-wise RMSD results for mean-peak-ADC values on HGCROC board 21 across multiple CALIB values. The RMSD values are derived from the sample distributions on each channel. The top figure shows link 0, while the bottom figure link 1.



Figure 33: Plot of HR channel-wise RMSD results for mean-peak-ADC values on HGCROC board 21 across multiple CALIB values. The RMSD values are derived from the sample distributions on each channel. The top figure shows link 0, while the bottom figure link 1. We note the spike in RMSD on channel 19, CALIB 512.

F Pedestal-alignment RMSD threshold

While the 10-ADC RMSD threshold has proven successful in catching misaligned pedestals on board 25, it has failed to notice the less-drastic misalignment in HGCROC-boards 26, 28 and 29. As can be seen in Figure 34, the mean-pedestal on channel 71 lags behind the rest by around 25 ADC units. Similar results can be found for boards 28 and 29 on channels 35 and 43, respectively. These misaligned channels are likely the result of the alignment procedure being inaccurate or inefficient, contrary to the ‘dead’ channel on board 25.

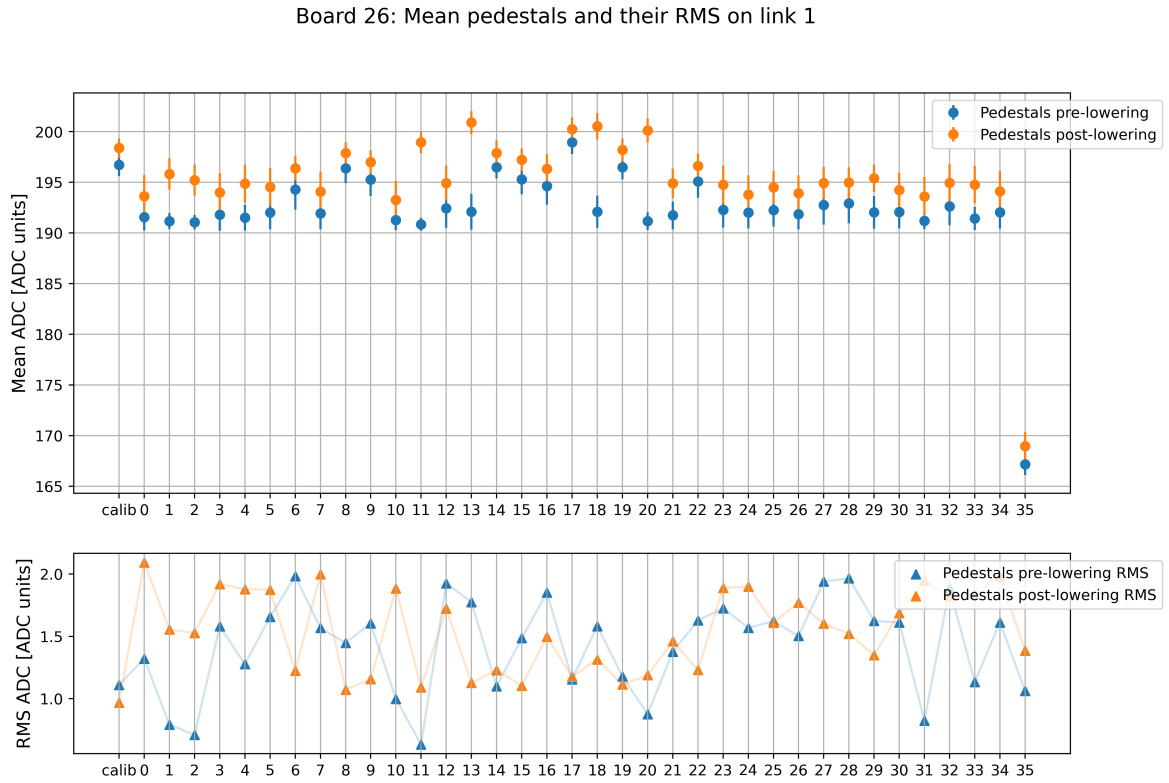


Figure 34: Plot of mean-pedestals across link 1 on HGCROC-board 26, before and after lowering. The means are taken over a 100 samples on each channel. The top figure shows the mean-pedestals with channel-wise RMSD-error-bars, while the bottom figure shows the channel-wise RMSD values. We note the misalignment between channel 71 and the rest of the link.

G TOA/TOT triggers and misfires

When the TOT threshold of around 950 ADC units is reached during a charge-injection, the ADC values between the TOA and TOT thresholds are reduced to -1, as seen in Figure 35. Notably, this is a unique feature of the time-scans - CALIB scans, for example, are entirely capable of capturing ADC and TDC values at the same time. Under the mistaken assumption that this issue only pertains HR time-scans and the TOT-TDC, we repeated the HR scans with the TOT-TDC turned off and TOA-TDC on. This resulted in pre-CC and HR time-scans suffering specific TDC-trigger issues - pre-CC having both TDCs active during testing, and HR only the TOA-TDC.

The pre-CC time-scan data on CALIB 4095 retains high link-wise RMSD values due to TOA/TOT-triggers, which fully overwrite the ADC-peak values. Taking a closer look at the pre-CC data, it is clear that only a small number of channels contain injections high enough to pass their TOT-threshold (see Section 5.3.2). However, an irregular response on even a single channel can result in a high link-wise RMSD. This issue is corroborated for boards: 21, 22, 24, 25, 29 and 30.

Due to only the TOA-TDC being active, the HR data contains less numerous, but stranger ADC-value irregularities. As can be seen in Figure 36, most of the peak-ADC values are no longer being overwritten, the TOT-TDC having been switched off. However, some samples still act as though they did not get the memo, resulting in a row of points at -1 ADC-units. What is harder to explain is the second row, a seeming continuation of the pedestal, which would not normally result from a TOA/TOT-trigger. It is likely that this anomalous behaviour is caused by the `pftool` time-scan script not having been written to account for switching-off only one TDC.

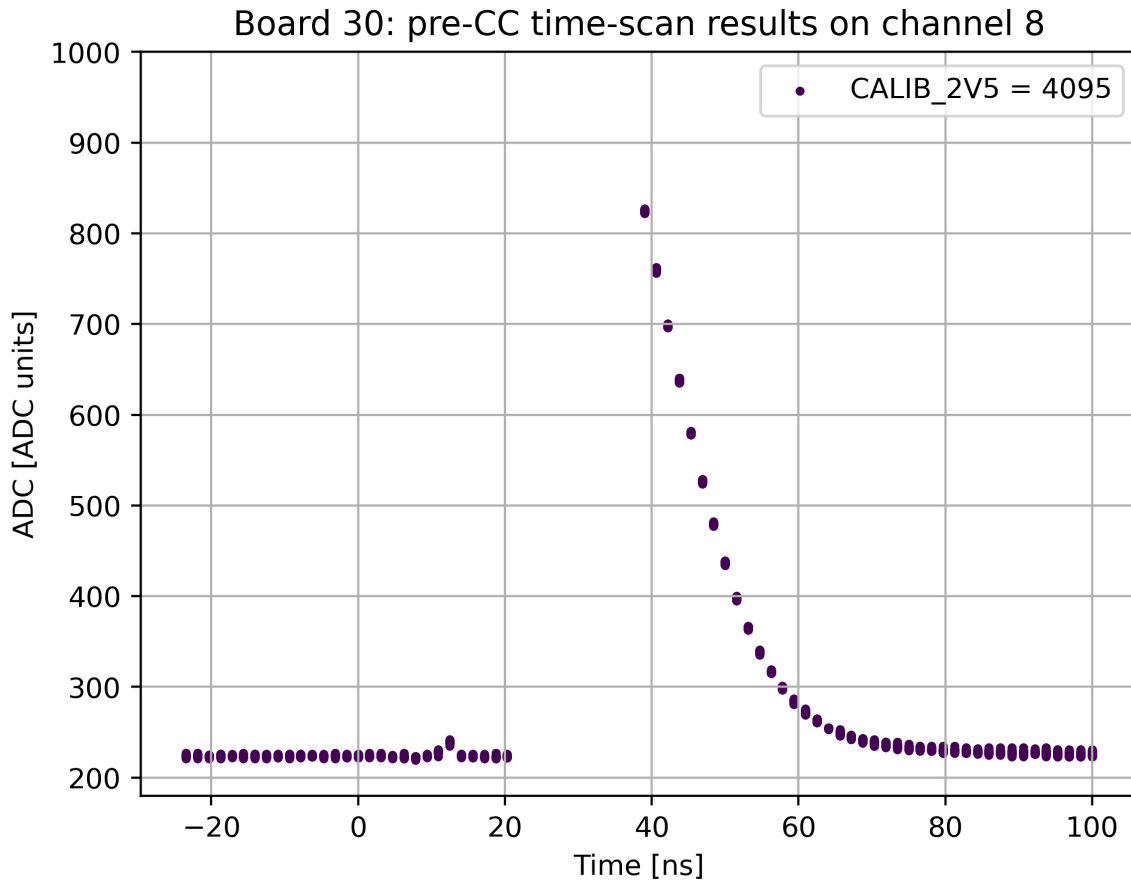


Figure 35: Plot of a pre-CC time-scan response on HGCROC board 30. The figure includes 50 samples per phase, taken at CALIB 4095 on channel 8. The seeming lack of ADC values between around 21-38 ns is caused by a TOT-trigger, which overwrites the ADC value to -1 in the TOT-active region.

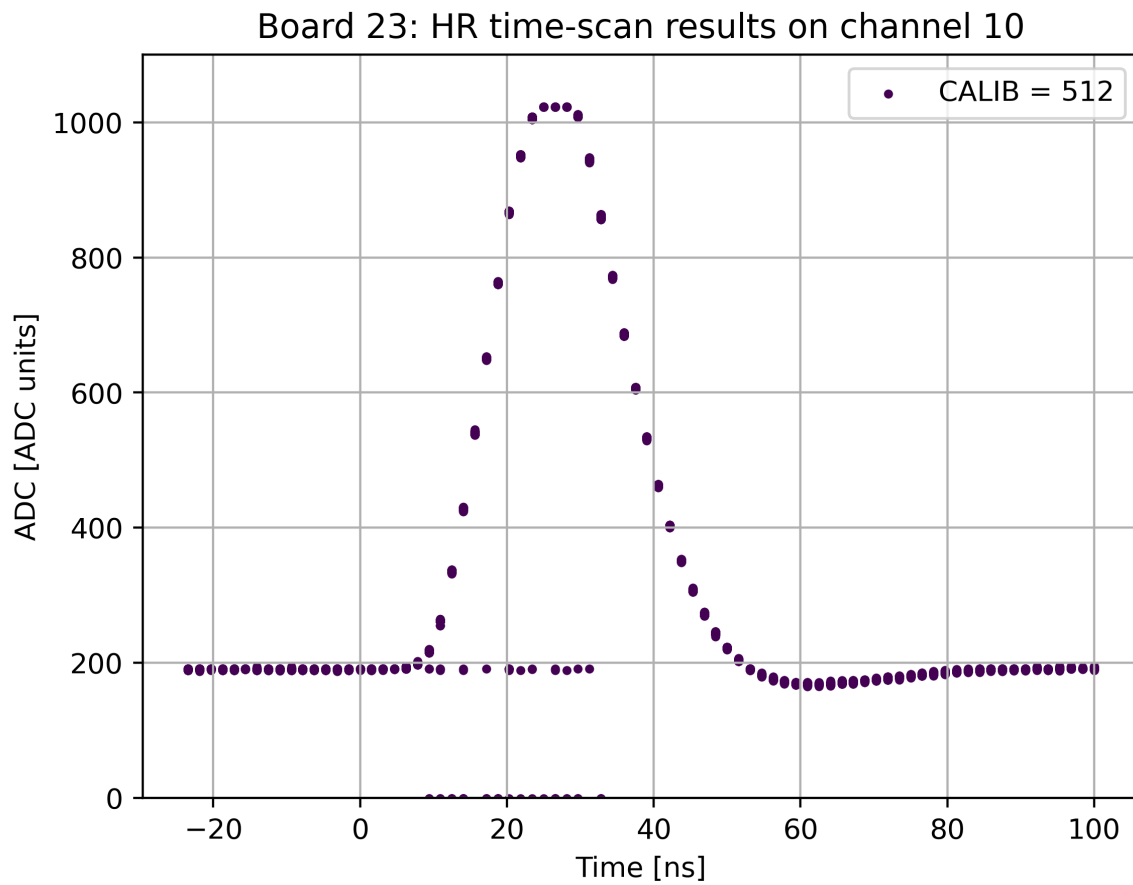


Figure 36: Plot of an HR time-scan response on HGCROC board 23. The figure includes 50 samples per phase, taken at CALIB 512 on channel 10. The plot shows two anomalous groups of samples. One of the groups looks like a continuation of the pedestal, while the other seems to be at -1 ADC units, indicating TOT-trigger. These irregularities are likely a result of switching-off the TOT-TDC, while leaving the TOA-TDC on.

H Identification algorithm plot examples

The figures below are some of the plots that can be made using functions of the identification algorithm described in Section 6.3.

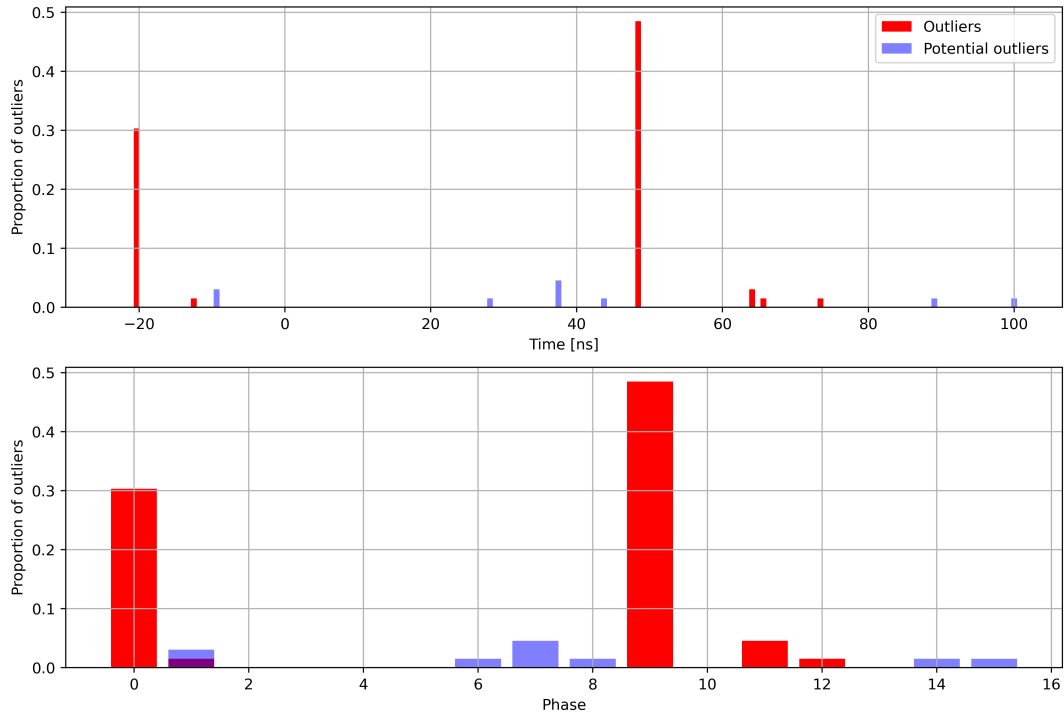


Figure 37: Example of a phase analysis plot. The data is taken from a pre-CC time-scan on board 20, taken with 50 samples, at CALIB 4095 on channel 0. The plot shows the proportion of outlier ADC samples found at specific time-points (top plot) or phases (bottom plot).

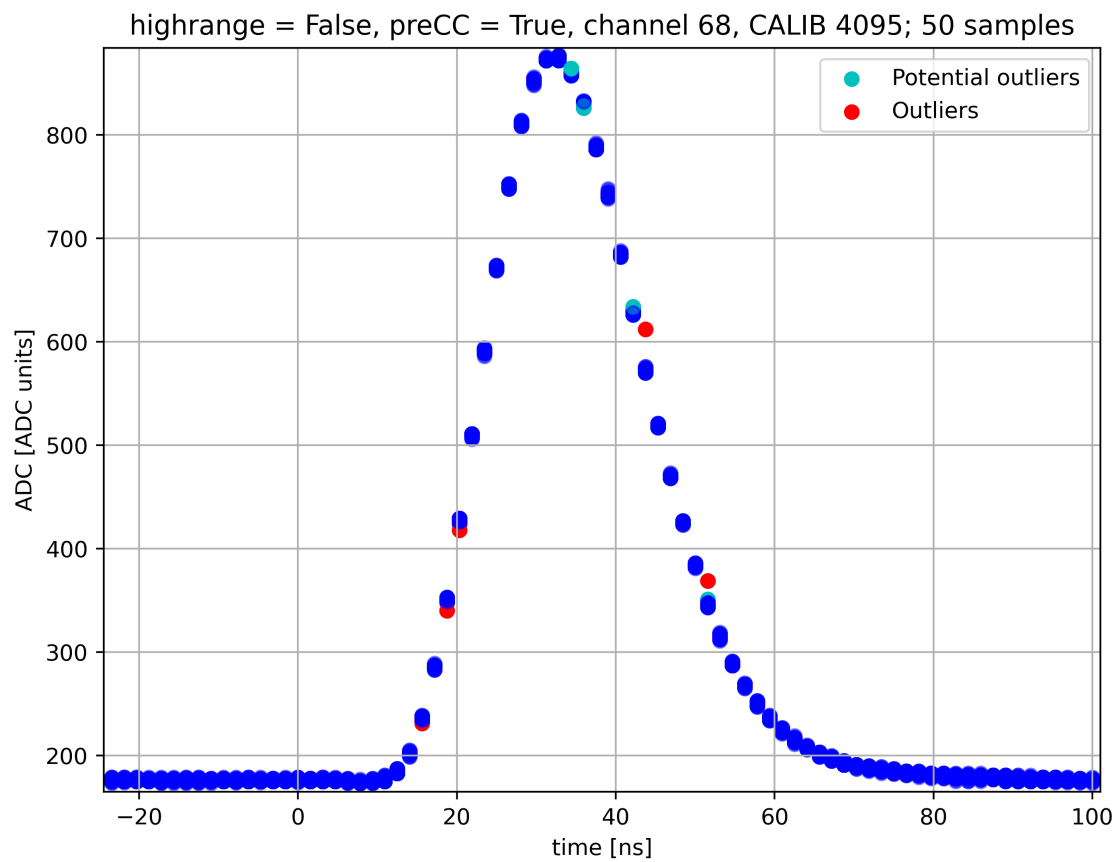


Figure 38: Example of a cluster-based identification plot. The data is taken from a pre-CC time-scan on board 20, taken with 50 samples, at CALIB 4095 on channel 68. Outlier samples are highlighted in red and potential outliers in cyan.

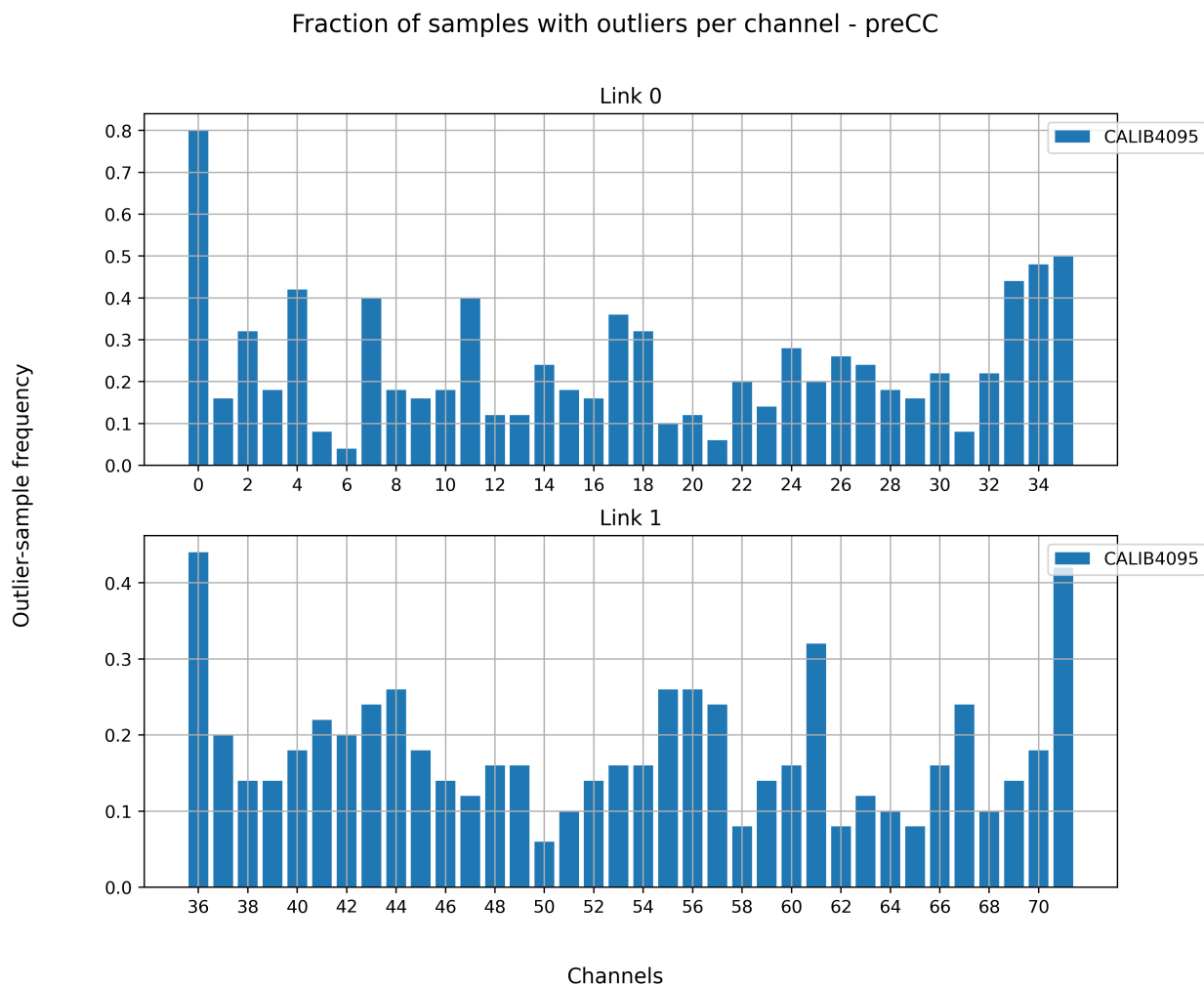


Figure 39: Example of a board-wide analysis plot. The data is taken from a pre-CC time-scan on board 20, taken with 50 samples, at CALIB 4095 on all channels. The outlier-sample frequency refers to the proportion of samples containing any identified outliers, excluding potential outliers, which are plotted separately.